Multiple testing for the topology of financial networks

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Abstract

This paper develops finite-sample multiple-testing procedures for inference on the topology of Diebold–Yilmaz connectedness networks. Connectedness matrices based on forecast error variance decompositions are typically dense, and empirical applications often rely on informal denoising rules to retain only the largest estimated spillovers. We provide resampling-based single-step and step-down procedures for testing pairwise directional connectedness while controlling the familywise error rate. The procedures are based on a restricted-complete-null permutation reference distribution for the maximum connectedness statistic and yield Monte Carlo adjusted p-values with strong finite-sample control under the stated conditions. We further develop induced tests for aggregate connectedness measures and extend the approach to networks with prespecified groups of nodes. Simulation experiments assess error control and power in finite samples, and empirical applications to global equity-return and volatility networks illustrate the use of the methods for denoising connectedness networks and comparing changes in network topology across crisis periods.

JEL classification: C12, C15, C32, G15.

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1 Introduction

Econometric measures of network connectedness are widely used to study the transmission of shocks across economic and financial systems. A leading example is the Diebold and Yilmaz (2009, 2012, 2014, hereafter DY) connectedness framework, which interprets forecast error variance decompositions (FEVDs) from a vector autoregression (VAR) as a weighted directed network. The resulting connectedness matrix provides estimates of the strength and direction of spillovers among variables. However, the usual implementation is essentially descriptive: estimated links are treated as features of the network without formal tests of whether they are statistically different from zero.

This lack of inference is consequential. The connectedness matrix is typically dense in finite samples, partly because sampling variation assigns positive estimated connectedness to many links. Empirical researchers therefore often rely on informal denoising rules, such as retaining only the largest estimated spillovers. Such rules can be useful for visualization, but they do not control the probability of retaining spurious links. More generally, without a multiple-testing framework, one cannot formally test hypotheses about the existence of individual links, aggregate spillovers, or group-level connectedness, nor can one construct statistically disciplined comparisons of network topology across periods.

Standard asymptotic inference does not readily provide such a framework, for several reasons. First, connectedness measures are nonnegative by construction and equal zero under the null hypothesis, placing the null on the boundary of the parameter space. Second, the mapping from VAR parameters to FEVD-based connectedness measures is nonlinear, complicating the derivation of analytic null distributions. Third, under the complete null, the distribution of the estimated connectedness measures depends on autoregressive nuisance parameters that are not specified by the null hypothesis. The resampling-based approach developed below avoids the need to derive analytic asymptotic null distributions. It constructs a restricted-complete-null permutation reference distribution for the maximum connectedness statistic, which is then used to obtain finite-sample multiplicity-adjusted p-values.

Existing work on inference for network connectedness is limited. Some contributions test the parameters of the underlying VAR rather than conducting inference on the network directly; others use bootstrap methods to construct confidence intervals for spillover statistics. For example, Claeys and Vašíček (2014) study contagion through structural breaks in VAR parameters, while Greenwood-Nimmo et al. (2019) and Bostanci and Yilmaz (2020) construct bootstrap confidence intervals for spillover measures, and Greenwood-Nimmo and Tarassow (2022) focus on scenario probabilities. Relatedly, Uematsu and Yamagata (2025) develop multiple-testing procedures for discovering network Granger causality in high-dimensional VARs. Their network is defined by nonzero VAR coefficients and their procedures target FDR control. In contrast, we conduct inference directly on DY connectedness measures, which are nonlinear functions of the VAR dynamics and the innovation covariance matrix, and we develop finite-sample procedures that control the familywise error rate (FWER) for the topology of the resulting FEVD-based network.

In this paper, we make three contributions. First, we develop single-step and step-down multiple-testing procedures for testing the statistical significance of pairwise directional connectedness measures. The procedures are based on a restricted-complete-null permutation reference distribution for the maximum connectedness statistic. This construction addresses the nuisance-parameter problem that arises under the complete null and accommodates the nonstandard feature that connectedness measures are nonnegative and lie on the boundary under the null. We use the Monte Carlo (MC) test methods of Dwass (1957), Barnard (1963), Birnbaum (1974), and Dufour (2006) to implement this permutation procedure without exhaustive enumeration and to obtain exact finite-sample adjusted p-values. Under the conditions stated below, the procedures control the FWER in the strong sense: the probability of falsely detecting at least one link is bounded by the nominal level, uniformly over configurations of true and false null hypotheses.

Second, we show how the adjusted pairwise p-values can be used to conduct valid inference on aggregate connectedness measures. These include the usual from-index, to-index, and aggregate spillover measures used in the DY framework. We also extend the methodology to settings in which the nodes are partitioned into prespecified groups. In this case, intra-group links are excluded from the tested family and from the resampling maxima used in the multiplicity adjustments. This allows researchers to focus inference on inter-group connectedness when within-group links are known, expected, or not of substantive interest, and it can improve power by reducing the multiplicity burden.

Third, we develop a framework for comparing statistically denoised network topologies across subsamples by classifying links as created, deleted, or unchanged according to their significance status. This separates link existence from link strength and provides a disciplined way to describe changes in statistically denoised network topology during crisis episodes. The framework therefore provides an interface between DY connectedness analysis and the large literature on contagion and changing transmission channels in financial networks.

We apply our method to the analysis of global equity market connectedness. In our first application, we revisit the equity return network from Diebold and Yilmaz (2009) using the authors' original data set, which covers 19 equity markets from January 10, 1992 to November 23, 2007. Our multiple testing procedure reveals previously unseen topological details. At the 1% FWER level, almost 40% (129/342) of pairwise connections are statistically insignificant, indicating considerable sparsity. Eliminating these insignificant connections materially changes the properties of the network. It reveals an underlying block structure with distinct groups of advanced markets, emerging Asian markets and Latin American markets. Furthermore, the Turkish market turns out to be a largely isolated node, receiving minimal inward spillovers and generating no outward spillover. Together, these changes reduce the spillover index by 4.5 percentage points, from 65.50% to 61.08%.

In our second application, we use a modern data set to compare statistically denoised G20 stock market volatility networks during the Global Financial Crisis (GFC) and the COVID crisis relative to pre-crisis benchmarks. Links are classified as created, deleted, or unchanged according to their significance status at the 1% FWER level. We show that spillover activity increases in both crises—which has already been widely documented—but the mechanisms generating this increase are very different.

During the GFC, the sparsity of the network increases relative to its pre-crisis state, as a large number of weak links are deleted, while a smaller number of stronger links are created. Spillovers originating in the North American and European markets are the dominant feature of the network pre-crisis and become even more important during the GFC, with the creation of 23 new links originating in this group compared to just 10 that are deleted. This reflects the origin of the GFC in the US financial markets and the substantial exposures of advanced economy financial institutions to the US securitized asset markets (Brunnermeier, 2009).

By contrast, during the COVID crisis, the volatility network becomes less sparse, with the creation of 55 links and the deletion of just 14. Unlike the GFC, new links are created originating from every region. Consequently, the increase in spillover activity is more broad-based, in the sense that it is attributed to a large number of comparatively weaker connections. This is likely a manifestation of the nature of the COVID crisis as a global public health emergency, the realized and anticipated effects of which were amplified by the financial system (Ramelli and Wagner, 2020), as well as the unprecedented restrictions on economic and social activities introduced by many governments around the world (Baker et al., 2020).

The remainder of the paper is organized as follows. Section 2 reviews the DY connectedness framework. Sections 3 and 4 develop the multiple-testing procedures for pairwise, aggregate, and group-based connectedness measures. Section 5 assesses error control and power in finite samples. Section 6 presents the empirical applications. Section 7 concludes. Proofs and additional results are collected in the online appendix.

2 Estimating connectedness

The DY framework uses a VAR model to estimate connectedness among N variables. Specifically, the vector $\mathbf{y}_t = (y_{1t}, \ldots, y_{Nt})'$ satisfies a covariance-stationary VAR(p)

$$\mathbf{y}_t = \mathbf{c} + \sum_{l=1}^{p} \mathbf{\Phi}_l \mathbf{y}_{t-l} + \boldsymbol{\varepsilon}_t, \quad t = 1, \ldots, T, \tag{1}$$

given initial conditions $\mathbf{y}_{-p+1}, \ldots, \mathbf{y}_0$. Here, the lag order p is treated as known, $\mathbf{c} = (c_1, \ldots, c_N)'$ is an $N \times 1$ intercept vector, and $\mathbf{\Phi}_l = [\phi_{ij,l}]$ for $l = 1, \ldots, p$ are $N \times N$ autoregressive coefficient matrices. The innovations $\{\boldsymbol{\varepsilon}_t\}_{t=1}^T$ are i.i.d. with $\mathbb{E}[\boldsymbol{\varepsilon}_t] = \mathbf{0}$ and $\mathbb{E}[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'] = \mathbf{\Sigma}$, where $\mathbf{\Sigma}$ is positive definite.

The moving average representation of (1) is $\mathbf{y}_t = \boldsymbol{\mu} + \sum_{l=0}^{\infty} \mathbf{\Psi}_l \boldsymbol{\varepsilon}_{t-l}$, with the mean vector given by $\boldsymbol{\mu} = (\mathbf{I}_N - \sum_{l=1}^{p} \mathbf{\Phi}_l)^{-1}\mathbf{c}$ and the $N \times N$ parameter matrices satisfying the recursion

$$\mathbf{\Psi}_l = \mathbf{\Phi}_1 \mathbf{\Psi}_{l-1} + \mathbf{\Phi}_2 \mathbf{\Psi}_{l-2} + \ldots + \mathbf{\Phi}_p \mathbf{\Psi}_{l-p}, \tag{2}$$

with $\mathbf{\Psi}_0 = \mathbf{I}_N$ and $\mathbf{\Psi}_l = \mathbf{0}$, for $l < 0$, where $\mathbf{I}_N$ is the $N \times N$ identity matrix. Following Diebold and Yilmaz (2014), we adopt the generalized forecast error variance decomposition (GFEVD) of Koop et al. (1996) and Pesaran and Shin (1998) as the foundation for network analysis. This has the advantage of being invariant to the ordering of the variables in the VAR model. The elements of the H-step-ahead GFEVD are given by

$$\theta_{ij}^H = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (\boldsymbol{\iota}_i' \mathbf{\Psi}_h \mathbf{\Sigma} \boldsymbol{\iota}_j)^2}{\sum_{h=0}^{H-1} (\boldsymbol{\iota}_i' \mathbf{\Psi}_h \mathbf{\Sigma} \mathbf{\Psi}_h' \boldsymbol{\iota}_i)}, \quad i, j = 1, \ldots, N, \tag{3}$$

where $\sigma_{jj} = \sigma_j^2$ is the variance of ε_{jt}, and $\boldsymbol{\iota}_i$ is an $N \times 1$ vector with a one in its i^{th} position and zeros elsewhere. The quantity $\theta_{ij}^H \geq 0$ in (3) represents the contribution to the total variance of y_i's H-step-ahead forecast error made by a one-standard-deviation shock to y_j.

The interpretation of GFEVDs is complicated by the fact that the sum $\sum_{j=1}^{N} \theta_{ij}^{H}$ may exceed 100% whenever $\mathbf{\Sigma}$ is non-diagonal. For this reason, Diebold and Yilmaz (2014) apply a normalization and measure the directional connectedness, or spillover, from y_j to y_i as

$$S_{i \leftarrow j}^{H} = \frac{\theta_{ij}^{H}}{\sum_{j=1}^{N} \theta_{ij}^{H}}, \quad i, j = 1, \ldots, N,$$

thereby ensuring that $\sum_{j=1}^{N} S_{i \leftarrow j}^{H} = 1$ and $\sum_{i,j=1}^{N} S_{i \leftarrow j}^{H} = N$. This normalization restores a proportional interpretation to the variance decomposition. A key insight of Diebold and Yilmaz (2014) is the recognition that cross-tabulating $S_{i \leftarrow j}^{H}$, as i, j range over $1, \ldots, N$, yields a weighted directed network capturing the pairwise relations among the system of N variables. The quantity $S_{i \leftarrow j}^{H}$ is the crucial building block to establish the topology of this network, as it represents the magnitude of the edge from node j to node i.

Another interpretation challenge arises when the N variables are structured into a set of $G \leq N$ groups and the $S_{i \leftarrow j}^{H}$ measures are aggregated according to this group structure. Indeed, aggregating these measures may lead to a situation where a group exhibits spillovers exceeding 100%. Following Greenwood-Nimmo et al. (2021), we therefore measure directional connectedness with the fractional value

$$C_{i \leftarrow j}^{H} = \frac{S_{i \leftarrow j}^{H}}{N}, \quad i, j = 1, \ldots, N, \tag{4}$$

wherein the denominator N equals $\sum_{i,j=1}^{N} S_{i \leftarrow j}^{H}$, the system-wide sum of directional connectedness measures. This renormalization ensures that $C_{i \leftarrow j}^{H} \times 100$ has a clear percentage interpretation even after organizing the variables into groups, because $\sum_{i,j=1}^{N} C_{i \leftarrow j}^{H} = 1$. Consequently, to facilitate understanding, the connectedness results in this paper are reported directly in percentages.

The information provided by $C_{i \leftarrow j}^{H}$ in (4) can be aggregated to measure other aspects of the network topology. Adapting definitions from Demirer et al. (2018), we define three measures to summarize directional connectedness within the network: (i) a "from index" (FIX) measure for the total directional connectedness *from all other nodes* to node i calculated as

$$C_{i \leftarrow \bullet}^{H} = \sum_{\substack{j=1 \\ j \neq i}}^{N} C_{i \leftarrow j}^{H}; \tag{5}$$

(ii) a "to index" (TIX) measure for the total directional connectedness *to all other nodes* from node j calculated as

$$C^H_{\bullet\leftarrow j} = \sum_{\substack{i=1 \\ i\neq j}}^{N} C^H_{i\leftarrow j}; \tag{6}$$

and (iii) an "aggregate index" (AIX) measure for the total directional connectedness across the entire network calculated as

$$C^H_{\bullet\leftarrow\bullet} = \sum_{\substack{i,j=1 \\ i\neq j}}^{N} C^H_{i\leftarrow j}. \tag{7}$$

Note that the division by N appearing in (4) renders these three definitions identical to those in Demirer et al. (2018). In the subsequent sections, we develop inferential procedures for the pairwise connectedness measures $C^H_{i\leftarrow j}$ in (4), as well as the aggregate connectedness measures defined in (5)–(7).

Because each equation in a VAR has the same right-hand-side regressors, equation-by-equation ordinary least squares (OLS) yields the same coefficient estimates as joint system estimation; see Hamilton (1994, Ch. 11). Indeed, the i^{th} row of (1) can be written as the linear regression

$$y_{it} = c_i + \sum_{l=1}^{p}\sum_{j=1}^{N} \phi_{ij,l}y_{j,t-l} + \varepsilon_{it}, \quad t = 1,\ldots,T,$$

wherein y_{it} depends on a constant term and p lags of each of the N variables in the system. Denote the OLS estimates as $\hat{\mathbf{c}} = (\hat{c}_1,\ldots,\hat{c}_N)'$ and $\widehat{\mathbf{\Phi}}_l = [\hat{\phi}_{ij,l}]_{N\times N}$, for $l = 1,\ldots,p$. These yield $\widehat{\mathbf{\Sigma}} = (1/T)\sum_{t=1}^{T}\hat{\boldsymbol{\varepsilon}}_t\hat{\boldsymbol{\varepsilon}}_t'$, where $\hat{\boldsymbol{\varepsilon}}_t = \mathbf{y}_t - \hat{\mathbf{c}} - \sum_{l=1}^{p}\widehat{\mathbf{\Phi}}_l\mathbf{y}_{t-l}$, and the estimates $\widehat{\mathbf{\Psi}}_0,\ldots,\widehat{\mathbf{\Psi}}_{H-1}$ of the corresponding moving average representation, defined in (2). In turn, the estimates $\hat{\theta}^H_{ij}$, $i,j = 1,\ldots,N$, of the GFEVD elements in (3) can then be computed, which yield the estimated node connectedness measures $\widehat{C}^H_{i\leftarrow j}$, as well as the estimated total directional connectedness measures $\widehat{C}^H_{i\leftarrow\bullet}$, $\widehat{C}^H_{\bullet\leftarrow j}$, and $\widehat{C}^H_{\bullet\leftarrow\bullet}$.

3 Testing connectedness

We wish to test whether $C^H_{i\leftarrow j} > 0$ for $i,j = 1,\ldots,N$, $i \neq j$. Formally, we assess node connectedness by testing the family of individual null hypotheses $\mathcal{H}_{i\not\leftarrow j}$ in the one-sided testing problem

$$\mathcal{H}_{i\not\leftarrow j} : C^H_{i\leftarrow j} = 0 \quad \text{versus} \quad \mathcal{H}_{i\leftarrow j} : C^H_{i\leftarrow j} > 0, \tag{8}$$

for $i, j = 1, \ldots, N$, $i \neq j$, while controlling the FWER. Throughout, $\mathcal{H}_{i \not\leftarrow j}$ denotes the null hypothesis that j does not spill over to i, while $\mathcal{H}_{i \leftarrow j}$ denotes the corresponding alternative.

To formalize the indexing, define the set of ordered pairs of distinct nodes

$$\mathcal{I}_M = \{(i, j) \in \{1, \ldots, N\}^2 : i \neq j\},$$

along with $M = |\mathcal{I}_M| = N^2 - N$. The restriction $i \neq j$ reflects that diagonal elements $C^H_{i \leftarrow i}$, which represent each node's own-variance contribution to its forecast error variance, are strictly positive by construction and are therefore excluded from the tested family. The remaining M ordered pairs are indexed as follows. Fix any bijection $\kappa_M : \mathcal{I}_M \to \mathcal{M} = \{1, \ldots, M\}$, and write its inverse as $\kappa_M^{-1}(\ell) = (i(\ell), j(\ell))$ for $\ell \in \mathcal{M}$. We then relabel the hypotheses via

$$\mathcal{H}_\ell = \mathcal{H}_{i(\ell) \not\leftarrow j(\ell)}, \quad \ell \in \mathcal{M}.$$

A convenient choice of κ_M is the lexicographic ordering of pairs (i, j) with i as the primary index, which induces

$$\mathcal{H}_1 = \mathcal{H}_{1 \not\leftarrow 2}, \ldots, \mathcal{H}_{N-1} = \mathcal{H}_{1 \not\leftarrow N},$$
$$\mathcal{H}_N = \mathcal{H}_{2 \not\leftarrow 1}, \ldots, \mathcal{H}_M = \mathcal{H}_{N \not\leftarrow N-1}.$$

Thus, each $\mathcal{H}_\ell$ is indexed by a unique ordered pair $(i(\ell), j(\ell)) \in \mathcal{I}_M$. In addition, define $\mathcal{M}_0 = \{\ell : \mathcal{H}_\ell \text{ is true}\}$ as the index set of true null hypotheses.

The FWER is formally defined as $\Pr\left(\text{Reject at least one } \mathcal{H}_\ell,\ \ell \in \mathcal{M}_0 \mid \bigcap_{\ell \in \mathcal{M}_0} \mathcal{H}_\ell\right)$. If all M null hypotheses are false, then the FWER is equal to zero by definition. Given the family of hypotheses, strong control of the FWER at (joint) level α requires that FWER $\leq \alpha$ for every possible choice $\mathcal{M}_0$. The estimated connectedness measures $\widehat{C}^H_\ell \geq 0$, $\ell = 1, \ldots, M$, serve as test statistics. For further reference, let the complete null hypothesis be written as

$$\mathcal{H}_{\bullet \not\leftarrow \bullet} : \bigcap_{\ell \in \mathcal{M}} \mathcal{H}_\ell, \tag{9}$$

which states that all M individual null hypotheses appearing in (8) are true.

Assumption 1 (Independence under the complete null)**.** *Under the complete null hypothesis $\mathcal{H}_{\bullet \not\leftarrow \bullet}$ in* (9)*, the components of $\boldsymbol{\varepsilon}_t$ are independent for each t.*

This restriction is imposed only when all pairwise null hypotheses in (8) are true and places no constraints on the model under the alternative. It is weaker than imposing Gaussian innovations with diagonal covariance under the complete null; it requires only cross-sectional independence of the innovation components and leaves their marginal distributions otherwise unrestricted.

Under $\mathcal{H}_{\bullet\not\leftarrow\bullet}$ and Assumption 1, the model reduces to

$$\mathbf{y}_t = \mathbf{c} + \sum_{l=1}^{p} \mathbf{D}_l \mathbf{y}_{t-l} + \varepsilon_t, \quad t = 1, \dots, T, \tag{10}$$

where each $\mathbf{D}_l$ is diagonal and $\mathbb{E}[\varepsilon_t \varepsilon_t'] = \mathbf{\Omega}$, with $\mathbf{\Omega}$ diagonal and positive definite. Hence (10) is a system of uncoupled univariate AR(p) equations with contemporaneously independent shocks.

In our framework, the distribution of $\widehat{C}_\ell^H$ under $\mathcal{H}_{\bullet\not\leftarrow\bullet}$ depends on the diagonal autoregressive coefficient matrices $\mathbf{D}_l$, $l = 1, \dots, p$, which are not specified by the complete null. We handle these nuisance parameters by constructing a restricted-complete-null reference distribution, which provides the basis for the finite-sample multiplicity adjustments developed below.

3.1 Bounding distribution

Although the restricted VAR in (10) imposes the complete null hypothesis $\mathcal{H}_{\bullet\not\leftarrow\bullet}$, it is not operational for testing purposes because $\mathbf{D}_l$, $l = 1, \dots, p$, are unknown. We therefore construct a bounding distribution based on the restricted complete null

$$\mathcal{H}^*_{\bullet\not\leftarrow\bullet} : \ \mathcal{H}_{\bullet\not\leftarrow\bullet} \text{ and } \{\mathbf{D}_l = \mathbf{D}_l^* : l = 1, \dots, p\}, \tag{11}$$

where $\mathbf{D}_l^* = [\phi^*_{ii,l}]$, $l = 1, \dots, p$, are $N \times N$ diagonal matrices chosen to be compatible with the complete null, so that $\mathcal{H}^*_{\bullet\not\leftarrow\bullet} \subseteq \mathcal{H}_{\bullet\not\leftarrow\bullet}$. We require that, for each $i = 1, \dots, N$, the roots of $1 - \sum_{l=1}^{p} \phi^*_{ii,l} z^l = 0$ lie outside the unit circle so that the moving-average representation is well-defined. In implementation, we set the nuisance parameters in (11) to the restricted OLS estimates from (10); i.e., $\mathbf{D}_l^* = \widehat{\mathbf{D}}_l$ for $l = 1, \dots, p$, where $\widehat{\mathbf{D}}_l$ is the diagonal matrix formed from the equation-by-equation OLS estimates of the lag-l autoregressive coefficients under (10).

We do not test $\mathcal{H}^*_{\bullet\not\leftarrow\bullet}$; it is a device used solely to construct a permutation reference distribution that will serve as a finite-sample bound for testing under the complete null $\mathcal{H}_{\bullet\not\leftarrow\bullet}$.

Under (11) the VAR becomes

$$\mathbf{y}_t = \mathbf{c} + \sum_{l=1}^{p} \mathbf{D}_l^* \, \mathbf{y}_{t-l} + \boldsymbol{\varepsilon}_t, \tag{12}$$

conditional on the pre-sample values $\mathbf{y}_{-p+1}, \ldots, \mathbf{y}_0$. By Assumption 1, for each t the components of $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \ldots, \varepsilon_{Nt})'$ are mutually independent, and for each i the sequence $\{\varepsilon_{it}\}_{t=1}^{T}$ is i.i.d. over time. Consequently, within each equation the innovations are *exchangeable*, which justifies generating simulated samples by permuting their order within equation.[1]

Denote the sample as $\mathbf{Y} = [\mathbf{Y}_1, \ldots, \mathbf{Y}_N]$, where $\mathbf{Y}_i = (y_{i,-p+1}, \ldots, y_{iT})'$, $i = 1, \ldots, N$. From the $\mathcal{H}^*_{\bullet\not\leftarrow\bullet}$–restricted innovations $\boldsymbol{\varepsilon}_t^* = \mathbf{y}_t - \sum_{l=1}^{p} \mathbf{D}_l^* \mathbf{y}_{t-l}$, $t = 1, \ldots, T$, construct a simulated sample $\widetilde{\mathbf{Y}}$ as follows. For each equation $i = 1, \ldots, N$, sample without replacement within equation $\tilde{\varepsilon}_{i1}, \ldots, \tilde{\varepsilon}_{iT}$ from $\varepsilon_{i1}^*, \ldots, \varepsilon_{iT}^*$, and generate $\widetilde{\mathbf{Y}}_i = (\tilde{y}_{i,-p+1}, \ldots, \tilde{y}_{iT})'$ recursively via

$$\tilde{y}_{it} = \sum_{l=1}^{p} \phi_{ii,l}^* \, \tilde{y}_{i,t-l} + \tilde{\varepsilon}_{it}, \tag{13}$$

for $t = 1, \ldots, T$, given the initial values $\tilde{y}_{i,-p+1} = y_{i,-p+1}, \ldots, \tilde{y}_{i0} = y_{i0}$. Collecting across equations yields $\widetilde{\mathbf{Y}} = [\widetilde{\mathbf{Y}}_1, \ldots, \widetilde{\mathbf{Y}}_N]$.[2]

Proposition 3.1. *Assume* $\mathbf{Y}$ *follows the VAR in* (1). *Let* $\tilde{m} = \max(\widetilde{C}_\ell^H : \ell \in \mathcal{M})$, *where the* $\widetilde{C}_\ell^H$ *are computed from a simulated sample* $\widetilde{\mathbf{Y}}$ *generated by* (13). *Under* $\mathcal{H}^*_{\bullet\not\leftarrow\bullet}$ *and Assumption 1, all within-equation permutations of the restricted innovations are equally likely. Hence the permutation distribution of* $\tilde{m}$ *based on the* $(T!)^N$ *permutations provides an exact finite-sample reference distribution for the observed statistic* $\hat{m} = \max(\widehat{C}_\ell^H : \ell \in \mathcal{M})$ *computed from* $\mathbf{Y}$.

Under the conditions of Proposition 3.1, $\hat{m}$ is pivotal, meaning its null permutation distribution does not depend on any nuisance parameters. In principle, exact critical values could be obtained from the permutation distribution based on all $(T!)^N$ equally likely within-equation permutations. In practice, we draw $B - 1$ permutations at random and use the finite-sample MC rank construction described below; see Dufour (2006).

In our context, this technique proceeds by generating $B - 1$ permuted samples $\widetilde{\mathbf{Y}}_b$, $b = 1, \ldots, B-1$, each via (13). With each such sample, the unrestricted VAR in (1) is estimated by OLS to obtain $\tilde{\mathbf{c}}_b$, $\widetilde{\boldsymbol{\Phi}}_{1,b}, \ldots, \widetilde{\boldsymbol{\Phi}}_{p,b}$, and $\widetilde{\boldsymbol{\Sigma}}_b$. These are then used to compute $\widetilde{C}_{\ell,b}^H$, $\ell = 1, \ldots, M$.

[1]A finite sequence is exchangeable if its joint distribution is invariant under permutations: for any permutation $(s_{i1}, \ldots, s_{iT})$ of $\{1, \ldots, T\}$, $(\varepsilon_{i1}, \ldots, \varepsilon_{iT}) \overset{d}{=} (\varepsilon_{i,s_{i1}}, \ldots, \varepsilon_{i,s_{iT}})$. With i.i.d. errors this holds trivially; permutation is equivalent to sampling without replacement. See, e.g., Randles and Wolfe (1979, Definition 1.3.6).

[2]The resampling scheme in (13) does not depend on the intercept $\mathbf{c}$ or the innovation variances. Because $\mathbf{c}$ is absorbed into $\boldsymbol{\varepsilon}_t^*$, the components of $\boldsymbol{\varepsilon}_t^*$ need not have zero mean. Permuting these implied error terms within each equation preserves their sample means and variances.

The simulated connectedness measures yield $\tilde{m}_b = \max(\widetilde{C}^H_{\ell,b} : \ell \in \mathcal{M})$, $b = 1, \ldots, B-1$. In what follows, we write $\mathbb{1}\{\cdot\}$ for the indicator function.

Proposition 3.2. *Assume that* $\mathbf{Y}$ *satisfies the VAR model in* (1) *and consider the statistic* $\hat{m} = \max(\widehat{C}^H_\ell : \ell \in \mathcal{M})$*. Define its MC p-value as*

$$\tilde{p}(\hat{m}) = \frac{B - R(\hat{m}) + 1}{B},$$

where $R(\hat{m}) = 1 + \sum_{b=1}^{B-1} \mathbb{1}\{\hat{m} > \tilde{m}_b\}$ *is the rank of* $\hat{m}$ *among* $\{\tilde{m}_1, \ldots, \tilde{m}_{B-1}, \hat{m}\}$*. If* αB *is an integer and there are no ties so that* $R(\hat{m})$ *is uniquely defined, then* $\Pr\left(\tilde{p}(\hat{m}) \le \alpha\right) = \alpha$*, under* $\mathcal{H}^*_{\bullet \not\leftarrow \bullet}$ *in* (11) *and the independence condition in Assumption 1.*[3]

Proposition 3.2 establishes that the max test which rejects when $\tilde{p}(\hat{m}) \le \alpha$ has size α in finite samples under $\mathcal{H}^*_{\bullet \not\leftarrow \bullet}$. This MC implementation of the permutation reference distribution is used to construct the multiplicity-adjusted p-values developed next.

The following assumption states the least-favourable bounding condition used to extend the exact restricted-complete-null result of Proposition 3.2 to strong FWER control under partial-null configurations. Let $\tilde{p}_{\mathcal{P}}(\widehat{C}^H_\ell)$ denote the multiplicity-adjusted MC p-value assigned to hypothesis $\mathcal{H}_\ell$ by procedure $\mathcal{P}$.

Assumption 2 (Least-favourable restricted complete null)**.** *For each multiple-testing procedure* $\mathcal{P}$ *considered below, for every non-empty* $\mathcal{M}_0 \subseteq \mathcal{M}$ *and every* $\alpha \in (0,1)$*,*

$$\Pr\left(\min_{\ell \in \mathcal{M}_0} \tilde{p}_{\mathcal{P}}(\widehat{C}^H_\ell) \le \alpha \;\middle|\; \bigcap_{\ell \in \mathcal{M}_0} \mathcal{H}_\ell\right) \le \Pr\left(\min_{\ell \in \mathcal{M}} \tilde{p}_{\mathcal{P}}(\widehat{C}^H_\ell) \le \alpha \;\middle|\; \mathcal{H}^*_{\bullet \not\leftarrow \bullet}\right).$$

This assumption states that the probability of at least one spurious rejection among any subset of true null hypotheses is bounded above by the corresponding full-family rejection probability under the restricted complete-null reference distribution. It is motivated by two structural features of the setting. First, the right-hand side of the inequality in Assumption 2 is based on the full-family rejection event, whereas the left-hand side involves only the true-null subset $\mathcal{M}_0 \subseteq \mathcal{M}$. Second, under any partial-null configuration, the connectedness measures corresponding to the true-null pairs are zero in population, so any rejection of these hypotheses arises solely from finite-sample variation; however, because false-null links may influence this variation through the nonlinear GFEVD mapping, Assumption 2 treats the

[3] Ties can be handled by replacing $R(\hat{m})$ by a lexicographic (tie-breaking) rank when computing the MC p-value, and exactness then follows from Proposition 2.4 in Dufour (2006).

restricted complete-null reference distribution as a conservative bound rather than an exact characterisation.

Assumption 2 plays the same logical role as the condition that links the resampling reference distribution to partial-null configurations in resampling-based strong-control arguments. In Westfall and Young (1993), this role is played by subset pivotality, which requires that, for any subset of true null hypotheses, the joint null distribution of the corresponding test statistics be the same under the partial-null configuration as under the complete null. This equality-in-distribution requirement is more restrictive than the one-sided familywise rejection bound imposed in Assumption 2, which requires only that the restricted complete-null reference distribution be conservative for the familywise rejection event. A related complete-null bounding logic is used in Luger (2025), where subset pivotality is not available for pairwise correlations and the complete-null randomization distribution is used as a conservative reference. In the present setting, subset pivotality is likewise not available because the distribution of the true-null test statistics may depend on the configuration of false-null pairs through the nonlinear GFEVD mapping. Assumption 2 is therefore maintained as a sufficient least-favourable bounding condition rather than derived analytically, and the simulation experiments in Section 5 evaluate the finite-sample performance of the proposed procedures under partial-null designs.

3.2 Single-step adjustments

Westfall and Young (1993) propose several resampling-based methods to adjust p-values for multiplicity. Adjusted p-values are the smallest significance levels at which individual hypotheses are rejected under a given multiple-testing procedure.

A straightforward extension of Westfall-Young's *single-step (SS) maxT adjusted p-values* to our setting is

$$p_{\text{SS}}(\widehat{C}^H_\ell) = \Pr\left(\max_{k\in\mathcal{M}} \widetilde{C}^H_k \geq \widehat{C}^H_\ell \,\middle|\, \mathcal{H}^*_{\bullet\not\leftarrow\bullet}\right), \quad \ell = 1, \ldots, M, \tag{14}$$

where $\widetilde{C}^H_k$ denotes a generic resampled connectedness measure computed from a sample generated under $\mathcal{H}^*_{\bullet\not\leftarrow\bullet}$ in (11). Here, the restricted complete null is used solely to obtain an exact finite-sample permutation reference for the max statistic (Proposition 3.2), in contrast with the original Westfall–Young formulation, which proceeds under the complete null.

Algorithm 3.1 (Single-step adjusted p-values)**.**

1. With the actual data $\mathbf{Y}$, estimate the VAR in (1) by OLS to obtain the unrestricted

estimates $\hat{\mathbf{c}}$, $\widehat{\mathbf{\Phi}}_1, \ldots, \widehat{\mathbf{\Phi}}_p$, and $\widehat{\mathbf{\Sigma}}$. Next, compute $\widehat{C}_\ell^H$, $\ell = 1, \ldots, M$.

2. Still with the actual data $\mathbf{Y}$, estimate (10) equation by equation by OLS to obtain the restricted estimates $\widehat{\mathbf{D}}_l$, $l = 1, \ldots, p$.

3. Choose the number of resampling draws $B - 1$ so that αB is an integer, where $\alpha \in (0, 1)$ is the nominal FWER level.

4. For $b = 1, \ldots, B - 1$, repeat the following steps:

   (a) simulate a sample $\widetilde{\mathbf{Y}}_b$ according to (13) using $\mathbf{D}_l^* = \widehat{\mathbf{D}}_l$, $l = 1, \ldots, p$;

   (b) with $\widetilde{\mathbf{Y}}_b$, estimate the VAR in (1) by OLS and compute $\widetilde{C}_{\ell,b}^H$, $\ell = 1, \ldots, M$;

   (c) set $\tilde{m}_b = \max(\widetilde{C}_{\ell,b}^H : \ell \in \mathcal{M})$.

5. For $\ell = 1, \ldots, M$, compute the SS adjusted MC p-value of $\widehat{C}_\ell^H$ as

$$\tilde{p}_{\mathrm{SS}}(\widehat{C}_\ell^H) = \frac{B - R_{\mathrm{SS}}(\widehat{C}_\ell^H) + 1}{B},$$

   where $R_{\mathrm{SS}}(\widehat{C}_\ell^H) = 1 + \sum_{b=1}^{B-1} \mathbb{1}\{\widehat{C}_\ell^H > \tilde{m}_b\}$ is the rank of $\widehat{C}_\ell^H$ among $\{\tilde{m}_1, \ldots, \tilde{m}_{B-1}, \widehat{C}_\ell^H\}$.

Proposition 3.3. *Under the conditions of Proposition 3.2 and Assumption 2 applied to $\mathcal{P} = \mathrm{SS}$, assume Algorithm 3.1 is implemented with αB an integer and that, for each $\ell \in \mathcal{M}$, there are no ties among $\{\tilde{m}_1, \ldots, \tilde{m}_{B-1}, \widehat{C}_\ell^H\}$ so that $R_{SS}(\widehat{C}_\ell^H)$ is uniquely defined. Then Algorithm 3.1 controls the finite-sample FWER at level α in the strong sense; i.e.,*

$$\Pr\left(\text{Reject at least one } \mathcal{H}_\ell,\ \ell \in \mathcal{M}_0 \,\middle|\, \bigcap_{\ell \in \mathcal{M}_0} \mathcal{H}_\ell\right) \leq \alpha.$$

The logic underlying this finite-sample bound is akin to the bound MC test of Dufour and Khalaf (2002, Theorem 4.1); see also the related arguments in Dufour (1989) and Luger (2025). Using the indexing defined above, the adjusted p-values are first obtained for the relabeled family as $\tilde{p}_{\mathrm{SS}}(\widehat{C}_\ell^H)$, $\ell \in \mathcal{M}$. They are then returned to the original matrix indexing using the inverse mapping $\kappa_M^{-1}(\ell) = (i(\ell), j(\ell))$ by setting

$$\tilde{p}_{\mathrm{SS}}\big(\widehat{C}_{i(\ell)\leftarrow j(\ell)}^H\big) = \tilde{p}_{\mathrm{SS}}\big(\widehat{C}_\ell^H\big), \quad \ell \in \mathcal{M}.$$

For diagonal pairs, $\tilde{p}_{\mathrm{SS}}(\widehat{C}_{i\leftarrow i}^H)$ is left undefined, since diagonal terms are not included in the tested family.

3.3 Step-down adjustments

A limitation of the p-value adjustment in (14) is that all the p-values are adjusted according to the distribution of the largest connectedness measure. Potentially less conservative p-values may be obtained from step-down adjustments that result in smaller p-values, while retaining the same protection against Type I errors. The underlying idea is analogous to Holm's (1979) sequential refinement of the Bonferroni adjustment, which eliminates from consideration any null hypothesis that is rejected at a previous step.

With the estimated connectedness measures $\widehat{C}_1^H, \ldots, \widehat{C}_M^H$, let the ordered statistics have index values $\pi_1, \ldots, \pi_M$ so that $\widehat{C}_{\pi_1}^H \geq \widehat{C}_{\pi_2}^H \ldots \geq \widehat{C}_{\pi_M}^H$. In the present context, Westfall and Young's (1993) definition of *step-down (SD) maxT adjusted p-values* can be extended as follows:

$$
\begin{aligned}
p_{\mathrm{SD}}(\widehat{C}_{\pi_1}^H) &= \Pr\left(\max_{\ell=1,\ldots,M} \widetilde{C}_{\pi_\ell}^H \geq \widehat{C}_{\pi_1}^H \mid \mathcal{H}^*_{\bullet \nleftarrow \bullet} \right), \\
p_{\mathrm{SD}}(\widehat{C}_{\pi_2}^H) &= \max\left[p_{\mathrm{SD}}(\widehat{C}_{\pi_1}^H),\, \Pr\left(\max_{\ell=2,\ldots,M} \widetilde{C}_{\pi_\ell}^H \geq \widehat{C}_{\pi_2}^H \mid \mathcal{H}^*_{\bullet \nleftarrow \bullet} \right) \right], \\
&\vdots \\
p_{\mathrm{SD}}(\widehat{C}_{\pi_r}^H) &= \max\left[p_{\mathrm{SD}}(\widehat{C}_{\pi_{r-1}}^H),\, \Pr\left(\max_{\ell=r,\ldots,M} \widetilde{C}_{\pi_\ell}^H \geq \widehat{C}_{\pi_r}^H \mid \mathcal{H}^*_{\bullet \nleftarrow \bullet} \right) \right], \\
&\vdots \\
p_{\mathrm{SD}}(\widehat{C}_{\pi_M}^H) &= \max\left[p_{\mathrm{SD}}(\widehat{C}_{\pi_{M-1}}^H),\, \Pr\left(\widetilde{C}_{\pi_M}^H \geq \widehat{C}_{\pi_M}^H \mid \mathcal{H}^*_{\bullet \nleftarrow \bullet} \right) \right],
\end{aligned}
\tag{15}
$$

with the sequence of index values $\pi_1, \ldots, \pi_M$ held *fixed* throughout. As in the single-step case, we condition on the restricted complete null $\mathcal{H}^*_{\bullet \nleftarrow \bullet}$ (rather than the complete null) to obtain an exact finite-sample permutation reference.

Instead of adjusting all the p-values according to the distribution of the maximum connectedness measure, this approach only adjusts the p-value of $\widehat{C}_{\pi_1}^H$ using this distribution. The remaining p-values are then adjusted according to more and more restricted sets of maximum connectedness measures. Note that the use of the max operator (outside the square brackets) guarantees that the SD adjusted p-values have the same step-down monotonicity as the original statistics; i.e., smaller p-values are associated with larger values of the $\widehat{C}_\ell^H$ test statistics. Power improvements can be expected with this approach as the SD adjusted p-values are uniformly equal to or smaller than their SS counterparts.

Extending Westfall and Young (1993, Algorithm 4.1, pp. 116–117) and Ge et al. (2003, Box 2) to our context, Algorithm 3.2 can be used to compute the MC version of the SD p-values defined in (15).

Algorithm 3.2 (Step-down adjusted p-values)**.**

1. With the actual data $\mathbf{Y}$, estimate the VAR in (1) by OLS to obtain the unrestricted estimates $\hat{\mathbf{c}}$, $\widehat{\boldsymbol{\Phi}}_1, \dots, \widehat{\boldsymbol{\Phi}}_p$, and $\widehat{\boldsymbol{\Sigma}}$. Next, compute $\widehat{C}^H_\ell$, $\ell = 1, \dots, M$.

2. Determine the index values $\pi_1, \dots, \pi_M$ that define the ordering $\widehat{C}^H_{\pi_1} \geq \widehat{C}^H_{\pi_2} \geq \cdots \geq \widehat{C}^H_{\pi_M}$.

3. Still with the actual data $\mathbf{Y}$, estimate (10) equation by equation by OLS to obtain the restricted estimates $\widehat{\mathbf{D}}_l$, $l = 1, \dots, p$.

4. Choose the number of resampling draws $B - 1$ so that αB is an integer, where $\alpha \in (0, 1)$ is the nominal FWER level.

5. For $b = 1, \dots, B - 1$, repeat the following steps:

   (a) simulate a sample $\widetilde{\mathbf{Y}}_b$ according to (13) using $\mathbf{D}^*_l = \widehat{\mathbf{D}}_l$, $l = 1, \dots, p$;

   (b) with $\widetilde{\mathbf{Y}}_b$, estimate the VAR in (1) by OLS and compute $\widetilde{C}^H_{\ell,b}$, $\ell = 1, \dots, M$;

   (c) compute the simulated step-down maxima as

$$\begin{aligned} \tilde{m}_{M,b} &= \widetilde{C}^H_{\pi_M,b}, \\ \tilde{m}_{\ell,b} &= \max\left(\tilde{m}_{\ell+1,b}, \widetilde{C}^H_{\pi_\ell,b}\right), \quad \ell = M-1, \dots, 1. \end{aligned}$$

6. For $\ell = 1, \dots, M$, compute the SD adjusted MC p-value of $\widehat{C}^H_{\pi_\ell}$ as

$$\tilde{p}_{\mathrm{SD}}\big(\widehat{C}^H_{\pi_\ell}\big) = \frac{B - R_{\mathrm{SD}}\big(\widehat{C}^H_{\pi_\ell}\big) + 1}{B},$$

   where $R_{\mathrm{SD}}\big(\widehat{C}^H_{\pi_\ell}\big) = 1 + \sum_{b=1}^{B-1} \mathbb{1}\big\{\widehat{C}^H_{\pi_\ell} > \tilde{m}_{\ell,b}\big\}$ is the rank of $\widehat{C}^H_{\pi_\ell}$ among $\{\tilde{m}_{\ell,1}, \dots, \tilde{m}_{\ell,B-1}, \widehat{C}^H_{\pi_\ell}\}$.

7. Enforce monotonicity by setting, for $\ell = 2, \dots, M$,

$$\tilde{p}_{\mathrm{SD}}\big(\widehat{C}^H_{\pi_\ell}\big) \leftarrow \max\left(\tilde{p}_{\mathrm{SD}}\big(\widehat{C}^H_{\pi_{\ell-1}}\big), \tilde{p}_{\mathrm{SD}}\big(\widehat{C}^H_{\pi_\ell}\big)\right),$$

   leaving $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_1})$ unchanged.

8. Recover $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_\ell)$, $\ell = 1, \dots, M$, by applying the inverse permutation induced by $\pi_1, \dots, \pi_M$; that is, if $\pi_r = \ell$ then set $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_\ell) = \tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_r})$.

Note that Step 7 applies the usual running-maximum correction to ensure that the MC step-down adjusted p-values are monotone with respect to the ordering $\widehat{C}^H_{\pi_1} \geq \cdots \geq \widehat{C}^H_{\pi_M}$.

Proposition 3.4. *Under the conditions of Proposition 3.2 and Assumption 2 applied to $\mathcal{P} = \mathrm{SD}$, assume Algorithm 3.2 is implemented with αB an integer and that, for each $\ell = 1, \ldots, M$, there are no ties among $\{\tilde{m}_{\ell,1}, \ldots, \tilde{m}_{\ell,B-1}, \widehat{C}^H_{\pi_\ell}\}$ so that $R_{SD}(\widehat{C}^H_{\pi_\ell})$ is uniquely defined. Then Algorithm 3.2 yields step-down adjusted MC p-values that control the finite-sample FWER at level α in the strong sense.*

Using the inverse mapping $\kappa_M^{-1}(\ell) = (i(\ell), j(\ell))$, the step-down adjusted p-values $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_\ell)$, $\ell \in \mathcal{M}$, are returned to the original matrix indexing via $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{i(\ell)\leftarrow j(\ell)}) = \tilde{p}_{\mathrm{SD}}(\widehat{C}^H_\ell)$. No adjusted p-values are assigned to diagonal terms, since these terms are not included in the tested family.

3.4 Aggregated connectedness

Let $\tilde{p}_{\mathcal{P}}(\widehat{C}^H_{i\leftarrow j})$ denote the multiplicity-adjusted p-value for $\widehat{C}^H_{i\leftarrow j}$ computed by procedure $\mathcal{P}$, where $\mathcal{P}$ is either the single-step procedure in Algorithm 3.1 or the step-down procedure in Algorithm 3.2. We now use these adjusted pairwise p-values to draw inference on aggregated connectedness measures.

FIX measures. We assess whether node i receives any spillovers by testing

$$\mathcal{H}_{i\not\leftarrow\bullet} : C^H_{i\leftarrow\bullet} = 0 \quad \text{versus} \quad \mathcal{H}_{i\leftarrow\bullet} : C^H_{i\leftarrow\bullet} > 0, \quad i = 1, \ldots, N, \tag{16}$$

with FWER control at level α. Because $C^H_{i\leftarrow j} \geq 0$, the null $\mathcal{H}_{i\not\leftarrow\bullet}$ holds if and only if all constituent pairwise nulls $\mathcal{H}_{i\not\leftarrow j}$, $j \neq i$, hold; equivalently,

$$\left\{C^H_{i\leftarrow\bullet} = 0\right\} \iff \left\{C^H_{i\leftarrow j} = 0 : \ j = 1, \ldots, N, \ j \neq i\right\}.$$

Accordingly, we define the induced p-value for $\widehat{C}^H_{i\leftarrow\bullet}$ as

$$\tilde{p}_{\mathcal{P}}(\widehat{C}^H_{i\leftarrow\bullet}) = \min\left\{\tilde{p}_{\mathcal{P}}(\widehat{C}^H_{i\leftarrow j}) : \ j = 1, \ldots, N, \ j \neq i\right\}, \tag{17}$$

so that $\mathcal{H}_{i\not\leftarrow\bullet}$ is rejected if and only if at least one constituent pairwise hypothesis is rejected (Savin, 1984). Let $\mathcal{M}_0^{\mathrm{FIX}} = \{i : \mathcal{H}_{i\not\leftarrow\bullet} \text{ is true}\}$. If $i \in \mathcal{M}_0^{\mathrm{FIX}}$, then all $\mathcal{H}_{i\not\leftarrow j}$, $j \neq i$, are true, so any rejection among the FIX hypotheses implies at least one rejection of a true pairwise null.

Therefore, by Propositions 3.3 and 3.4, the rejection rules $\tilde{p}_{\mathcal{P}}(\widehat{C}^{H}_{i\leftarrow\bullet}) \leq \alpha$, $i = 1, \ldots, N$, control the FWER at level α in the strong sense for the family of hypotheses in (16).

TIX measures. An identical argument applies to the to-indices in (6). We assess whether node j generates any spillovers by testing

$$\mathcal{H}_{\bullet\not\leftarrow j} : C^{H}_{\bullet\leftarrow j} = 0 \quad \text{versus} \quad \mathcal{H}_{\bullet\leftarrow j} : C^{H}_{\bullet\leftarrow j} > 0, \quad j = 1, \ldots, N,$$

using the induced p-values

$$\tilde{p}_{\mathcal{P}}(\widehat{C}^{H}_{\bullet\leftarrow j}) = \min\left\{\tilde{p}_{\mathcal{P}}(\widehat{C}^{H}_{i\leftarrow j}) : \ i = 1, \ldots, N, \ i \neq j\right\}, \quad j = 1, \ldots, N, \tag{18}$$

where $\widehat{C}^{H}_{\bullet\leftarrow j} = \sum_{i\neq j} \widehat{C}^{H}_{i\leftarrow j}$. Let $\mathcal{M}_0^{\text{TIX}} = \{j : \mathcal{H}_{\bullet\not\leftarrow j} \text{ is true}\}$. Then Propositions 3.3 and 3.4 imply

$$\Pr\left(\min_{j\in\mathcal{M}_0^{\text{TIX}}} \tilde{p}_{\mathcal{P}}(\widehat{C}^{H}_{\bullet\leftarrow j}) \leq \alpha \;\middle|\; \bigcap_{j\in\mathcal{M}_0^{\text{TIX}}} \mathcal{H}_{\bullet\not\leftarrow j}\right) \leq \alpha.$$

so the resulting tests control the TIX FWER at level α in the strong sense.

AIX measure. Finally, we assess whether the network exhibits any connectedness at all by testing

$$\mathcal{H}_{\bullet\not\leftarrow\bullet} : C^{H}_{\bullet\leftarrow\bullet} = 0 \quad \text{versus} \quad \mathcal{H}_{\bullet\leftarrow\bullet} : C^{H}_{\bullet\leftarrow\bullet} > 0,$$

where $\mathcal{H}_{\bullet\not\leftarrow\bullet}$ is the complete null in (9). We use the induced p-value

$$\tilde{p}_{\mathcal{P}}(\widehat{C}^{H}_{\bullet\leftarrow\bullet}) = \min\left\{\tilde{p}_{\mathcal{P}}(\widehat{C}^{H}_{i\leftarrow j}) : \ i, j = 1, \ldots, N, \ i \neq j\right\}. \tag{19}$$

Under $\mathcal{H}_{\bullet\not\leftarrow\bullet}$, rejecting this AIX null implies rejecting at least one true pairwise null, which occurs with probability at most α by Propositions 3.3 and 3.4. Hence the resulting test has level α.

4 Testing connectedness in the presence of groups

In this section, we extend the testing procedures in Section 3 to settings in which the nodes are partitioned into prespecified groups. Group structure restricts the family of hypotheses by excluding intra-group links, so that only inter-group connectedness is tested. This restriction may arise, for example, when within-group linkages are already well established or are not of substantive interest. In many applications, group structure is empirically plausible. For instance, financial institutions or markets may form clusters due to geographic proximity, trade ties, shared exposures, or common shocks. When reliable group information is available *a priori*, restricting attention to inter-group links can improve power by reducing the multiplicity burden.

Group membership is represented by sets $\mathcal{G}_1, \ldots, \mathcal{G}_G$, where $1 < G \leq N$ and $\mathcal{G}_g \subseteq \{1, \ldots, N\}$ collects the node indices in group g. We assume that (i) each node belongs to exactly one group, (ii) groups are disjoint, and (iii) groups may be singletons. When each node is treated as its own group, $G = N$. We reorder the variables so that $\mathbf{y}_t = (\mathbf{y}_t^{1\prime}, \ldots, \mathbf{y}_t^{G\prime})'$, where the vector $\mathbf{y}_t^g$ collects the time-t observations of the variables in group g. Let $g(i) \in \{1, \ldots, G\}$ denote the group index of node i, so that $i \in \mathcal{G}_{g(i)}$.

In the presence of groups, the individual null hypotheses remain the pairwise node-level nulls in (8), but the family over which multiplicity is controlled is restricted to inter-group links. Specifically, define the set of ordered inter-group pairs

$$\mathcal{I}_K = \big\{(i,j) \in \{1, \ldots, N\}^2 : i \neq j,\ g(i) \neq g(j)\big\}, \tag{20}$$

with

$$K = |\mathcal{I}_K| = N^2 - \sum_{g=1}^{G} |\mathcal{G}_g|^2.$$

We test

$$\mathcal{H}_{i \not\leftarrow j} : C^H_{i \leftarrow j} = 0 \quad \text{versus} \quad \mathcal{H}_{i \leftarrow j} : C^H_{i \leftarrow j} > 0, \qquad (i,j) \in \mathcal{I}_K, \tag{21}$$

while controlling the FWER over this restricted family. Thus, intra-group links are excluded from both the family of tested hypotheses and the resampling maxima used in the p-value adjustments. This reduction in the number of tested links is expected to yield power gains.

Let $\kappa_K : \mathcal{I}_K \to \mathcal{K} = \{1, \ldots, K\}$ be a bijection, with inverse $\kappa_K^{-1}(k) = (i(k), j(k))$, $k \in \mathcal{K}$. We then relabel the inter-group connectedness measures by

$$C^H_k = C^H_{i(k) \leftarrow j(k)}, \quad k \in \mathcal{K}, \tag{22}$$

so that intra-group connections are excluded by construction. With the relabeling $\mathcal{H}_k =$

$\mathcal{H}_{i(k)\not\leftarrow j(k)}$ for $k \in \mathcal{K}$, the complete null of no inter-group connectedness at the node level is

$$\mathcal{H}_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet} : \bigcap_{k \in \mathcal{K}} \mathcal{H}_k, \tag{23}$$

and we let $\mathcal{K}_0 = \{k : \mathcal{H}_k \text{ is true}\}$ denote the index set of true null hypotheses.

Assumption 3 (Cross-group independence under the inter-group complete null)**.** *Under the inter-group complete null hypothesis $\mathcal{H}_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet}$ in (23), the innovation subvectors $\varepsilon_t^1, \ldots, \varepsilon_t^G$ are independent for each t.*

This restriction is imposed only when all inter-group pairwise null hypotheses in (21) are true. It permits arbitrary dependence within each group and places no constraints on the model under the alternative. It is weaker than imposing Gaussian innovations with block-diagonal covariance under the inter-group complete null; it requires only cross-group independence of the innovation subvectors and leaves their within-group joint distributions otherwise unrestricted.

The hypothesis $\mathcal{H}_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet}$ can be represented by the G-block diagonal VAR specification

$$\mathbf{y}_t = \mathbf{c} + \sum_{l=1}^{p} \mathbf{B}_l \, \mathbf{y}_{t-l} + \varepsilon_t, \quad t = 1, \ldots, T, \tag{24}$$

where $\mathbf{B}_l = \mathrm{diag}(\mathbf{B}_l^1, \ldots, \mathbf{B}_l^G)$ and $\{\varepsilon_t\}_{t=1}^T$ are i.i.d. with $\mathbb{E}[\varepsilon_t] = \mathbf{0}$ and $\mathbb{E}[\varepsilon_t \varepsilon_t'] = \mathbf{\Omega}$, where $\mathbf{\Omega}$ is positive definite and block diagonal with blocks $(\mathbf{\Sigma}_1, \ldots, \mathbf{\Sigma}_G)$. For $g = 1, \ldots, G$, the matrices $\mathbf{B}_l^g$, $l = 1, \ldots, p$, and $\mathbf{\Sigma}_g$ have dimension $|\mathcal{G}_g| \times |\mathcal{G}_g|$.

As in Section 3, we construct a bounding distribution based on a point null,

$$\mathcal{H}^*_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet} : \mathcal{H}_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet} \text{ and } \{\mathbf{B}_l^g = \mathbf{B}_l^{g*} : l = 1, \ldots, p, \; g = 1, \ldots, G\}, \tag{25}$$

where $\mathbf{B}_l^{g*}$ are fixed values that ensure $\mathcal{H}^*_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet} \subseteq \mathcal{H}_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet}$ and yield a well-defined stationary moving-average representation within each block.

Denote the observed sample by $\mathbf{Y} = [\mathbf{Y}_1, \ldots, \mathbf{Y}_G]$, where $\mathbf{Y}_g = [\mathbf{y}^g_{-p+1}, \ldots, \mathbf{y}^g_T]'$, $g = 1, \ldots, G$. Moreover, let $\varepsilon_t^{g*} = \mathbf{y}_t^g - \sum_{l=1}^p \mathbf{B}_l^{g*} \mathbf{y}_{t-l}^g$, for $t = 1, \ldots, T$ and $g = 1, \ldots, G$, denote the innovations implied by imposing $\mathcal{H}^*_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet}$ in (25). A simulated sample is obtained as follows. For each group $g = 1, \ldots, G$, draw an independent random permutation π_g of $\{1, \ldots, T\}$ and set

$$\tilde{\varepsilon}_t^g = \varepsilon^{g*}_{\pi_g(t)}, \quad t = 1, \ldots, T.$$

That is, we permute the time indices separately for each group while keeping each innovation vector $\boldsymbol{\varepsilon}_t^{g*}$ intact, so that cross-sectional dependence within a group is preserved. We then construct the column block $\widetilde{\mathbf{Y}}_g = [\tilde{\mathbf{y}}_{-p+1}^g, \ldots, \tilde{\mathbf{y}}_T^g]'$ from the recursion

$$\tilde{\mathbf{y}}_t^g = \sum_{l=1}^{p} \mathbf{B}_l^{g*} \tilde{\mathbf{y}}_{t-l}^g + \tilde{\boldsymbol{\varepsilon}}_t^g, \quad t = 1, \ldots, T, \tag{26}$$

given the initial values $\tilde{\mathbf{y}}_0^g = \mathbf{y}_0^g, \ldots, \tilde{\mathbf{y}}_{-p+1}^g = \mathbf{y}_{-p+1}^g$. Collecting the column blocks together yields the simulated sample $\widetilde{\mathbf{Y}} = \big[\widetilde{\mathbf{Y}}_1, \ldots, \widetilde{\mathbf{Y}}_G\big]$.

4.1 p-value adjustments in the presence of groups

Here we detail the procedure for computing step-down adjusted p-values that control the FWER in the presence of groups. We focus on step-down adjustments; the corresponding single-step adjustments are obtained by replacing the step-down maxima below with the full-family maximum. The procedure is the grouped analogue of Algorithm 3.2 in Section 3.3.

Algorithm 4.1 (Step-down adjusted p-values in the presence of groups)**.**

1. With the observed data $\mathbf{Y}$, estimate the unrestricted VAR(p) in (1) by OLS to obtain $\hat{\mathbf{c}}$, $\widehat{\boldsymbol{\Phi}}_1, \ldots, \widehat{\boldsymbol{\Phi}}_p$, and $\widehat{\boldsymbol{\Sigma}}$. Next, compute the inter-group connectedness estimates $\widehat{C}_k^H$, $k = 1, \ldots, K$.

2. Determine the index values $\pi_1, \ldots, \pi_K$ that define the ordering $\widehat{C}_{\pi_1}^H \geq \widehat{C}_{\pi_2}^H \geq \cdots \geq \widehat{C}_{\pi_K}^H$.

3. Still with the observed data, estimate the restricted VAR in (24) by OLS under the block-diagonal restriction $\mathbf{B}_l = \mathrm{diag}(\mathbf{B}_l^1, \ldots, \mathbf{B}_l^G)$ to obtain $\widehat{\mathbf{B}}_l^g$, $l = 1, \ldots, p$, $g = 1, \ldots, G$.

4. Choose the number of resampling draws $B - 1$ so that αB is an integer, where $\alpha \in (0, 1)$ is the nominal FWER level.

5. For $b = 1, \ldots, B - 1$, repeat the following steps:

   (a) Simulate a sample $\widetilde{\mathbf{Y}}_b = [\widetilde{\mathbf{Y}}_{1,b}, \ldots, \widetilde{\mathbf{Y}}_{G,b}]$ according to (26) with $\mathbf{B}_l^{g*} = \widehat{\mathbf{B}}_l^g$, $l = 1, \ldots, p$, $g = 1, \ldots, G$.

   (b) With $\widetilde{\mathbf{Y}}_b$, estimate the unrestricted VAR in (1) by OLS and compute the simulated inter-group connectedness measures $\widetilde{C}_{k,b}^H$, $k = 1, \ldots, K$.

(c) Compute the simulated step-down maxima as

$$\tilde{m}_{K,b} = \widetilde{C}^H_{\pi_K,b},$$
$$\tilde{m}_{k,b} = \max\left(\tilde{m}_{k+1,b}, \widetilde{C}^H_{\pi_k,b}\right), \quad k = K-1, \ldots, 1.$$

6. For $k = 1, \ldots, K$, compute the SD adjusted MC p-value of $\widehat{C}^H_{\pi_k}$ as

$$\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_k}) = \frac{B - R_{\mathrm{SD}}(\widehat{C}^H_{\pi_k}) + 1}{B},$$

where $R_{\mathrm{SD}}(\widehat{C}^H_{\pi_k}) = 1 + \sum_{b=1}^{B-1} \mathbb{1}\{\widehat{C}^H_{\pi_k} > \tilde{m}_{k,b}\}$ is the rank of $\widehat{C}^H_{\pi_k}$ among $\{\tilde{m}_{k,1}, \ldots, \tilde{m}_{k,B-1}, \widehat{C}^H_{\pi_k}\}$.

7. Enforce monotonicity by setting, for $k = 2, \ldots, K$,

$$\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_k}) \leftarrow \max\left(\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_{k-1}}), \tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_k})\right),$$

leaving $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_1})$ unchanged.

8. Recover $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_k)$, $k = 1, \ldots, K$, by applying the inverse permutation induced by $\pi_1, \ldots, \pi_K$; that is, if $\pi_r = k$ then set $\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_k) = \tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_r})$.

The following assumption is the group-level analogue of Assumption 2. Let $\tilde{p}_{\mathcal{P}}(\widehat{C}^H_k)$ denote the multiplicity-adjusted MC p-value assigned to the inter-group hypothesis $\mathcal{H}_k$ by procedure $\mathcal{P}$, for $k \in \mathcal{K}$.

Assumption 4 (Least-favourable restricted complete null with groups)**.** *For each multiple-testing procedure $\mathcal{P}$ applied to the inter-group family, for every non-empty $\mathcal{K}_0 \subseteq \mathcal{K}$ and every $\alpha \in (0,1)$,*

$$\Pr\left(\min_{k\in\mathcal{K}_0} \tilde{p}_{\mathcal{P}}(\widehat{C}^H_k) \leq \alpha \;\middle|\; \bigcap_{k\in\mathcal{K}_0} \mathcal{H}_k\right) \leq \Pr\left(\min_{k\in\mathcal{K}} \tilde{p}_{\mathcal{P}}(\widehat{C}^H_k) \leq \alpha \;\middle|\; \mathcal{H}^*_{\mathcal{G}_\bullet \not\leftrightarrow \mathcal{G}_\bullet}\right).$$

Assumption 4 states that the probability of at least one spurious rejection among any subset of true inter-group null hypotheses is bounded above by the corresponding full-family rejection probability under the restricted inter-group complete null. It treats the restricted block-diagonal reference distribution as conservative for familywise rejection events involving true inter-group nulls, while allowing within-group dynamic linkages and within-group contemporaneous dependence.

Proposition 4.1. *Under $\mathcal{H}^*_{\mathcal{G}_\bullet \nleftarrow \mathcal{G}_\bullet}$ and Assumption 3, the group-wise permutation scheme in (26) provides an exact finite-sample reference distribution for the full-family maximum over $\mathcal{K}$. Under Assumption 4 applied to $\mathcal{P} = \mathrm{SD}$, assume Algorithm 4.1 is implemented with αB an integer and that, for each $k = 1, \dots, K$, there are no ties among $\{\tilde{m}_{k,1}, \dots, \tilde{m}_{k,B-1}, \widehat{C}^H_{\pi_k}\}$ so that $R_{\mathrm{SD}}(\widehat{C}^H_{\pi_k})$ is uniquely defined. Then Algorithm 4.1 controls the finite-sample FWER at level α in the strong sense for the family of node-level inter-group hypotheses $\{\mathcal{H}_k : k \in \mathcal{K}\}$.*

Though it would not be computationally efficient, replacing the step-down maxima $\tilde{m}_{k,b}$ in Step 5(c) by the common maximum $\tilde{m}_b = \max\{\widetilde{C}^H_{\pi_r,b} : r = 1, \dots, K\}$ for all k yields the single-step adjusted p-values. The same argument as in Proposition 4.1, with Assumption 4 applied to $\mathcal{P} = \mathrm{SS}$, gives strong finite-sample FWER control for the single-step version.[4]

In the following, let $\tilde{p}_{\mathcal{P}}(\widehat{C}^H_k)$ denote the multiplicity-adjusted p-value assigned to the inter-group hypothesis $\mathcal{H}_k$ by procedure $\mathcal{P}$, for $k \in \mathcal{K}$. Using the inverse mapping $\kappa_K^{-1}(k) = (i(k), j(k))$, the adjusted p-values are returned to the original matrix indexing via

$$\tilde{p}_{\mathcal{P}}\big(\widehat{C}^H_{i(k) \leftarrow j(k)}\big) = \tilde{p}_{\mathcal{P}}\big(\widehat{C}^H_k\big), \quad k \in \mathcal{K}.$$

For intra-group pairs (i, j) satisfying $g(i) = g(j)$, $\tilde{p}_{\mathcal{P}}(\widehat{C}^H_{i \leftarrow j})$ is left undefined, since such links are not included in the tested family.

4.2 Total connectedness in the presence of groups

In the presence of groups, the node-level aggregate indices FIX and TIX are modified to include only directional connectedness with nodes outside the node's own group. Let $\mathcal{O}(i) = \{1, \dots, N\} \setminus \mathcal{G}_{g(i)}$ denote the set of nodes outside the group containing node i. We define the node-level FIX and TIX measures in the presence of groups as, respectively,

$$C^H_{i \leftarrow \mathcal{O}(i)} = \sum_{j \in \mathcal{O}(i)} C^H_{i \leftarrow j}, \tag{27}$$

$$C^H_{\mathcal{O}(j) \leftarrow j} = \sum_{i \in \mathcal{O}(j)} C^H_{i \leftarrow j}. \tag{28}$$

For the grand total AIX in the presence of groups, we use

$$C^H_{\mathcal{G}} = \sum_{(i,j) \in \mathcal{I}_K} C^H_{i \leftarrow j}, \tag{29}$$

[4] Algorithm 3.2 is obtained as a special case of Algorithm 4.1 when each node is treated as a singleton group.

which represents total inter-group connectedness, where $\mathcal{I}_K$ is the set of ordered inter-group pairs defined in (20). If each node constitutes a singleton group, then $\mathcal{O}(i) = \{1, \ldots, N\} \setminus \{i\}$ and the measures in (27)–(29) reduce to those in (5)–(7).

Inference for (27)–(29) follows the induced-p-value approach of Section 3.4, now using the multiplicity-adjusted pairwise p-values $\tilde{p}_{\mathcal{P}}(\widehat{C}^H_{i\leftarrow j})$ for $(i,j) \in \mathcal{I}_K$. These p-values correspond to the node-level inter-group hypotheses included in the restricted family over which FWER is controlled. As in Section 3.4, because $C^H_{i\leftarrow j} \geq 0$, we have

$$C^H_{i\leftarrow\mathcal{O}(i)} = 0 \quad \Longleftrightarrow \quad C^H_{i\leftarrow j} = 0 \ \text{ for all } j \in \mathcal{O}(i).$$

Hence we test the N hypotheses

$$\mathcal{H}_{i\not\leftarrow\mathcal{O}(i)} : C^H_{i\leftarrow\mathcal{O}(i)} = 0 \quad \text{versus} \quad \mathcal{H}_{i\leftarrow\mathcal{O}(i)} : C^H_{i\leftarrow\mathcal{O}(i)} > 0, \quad i = 1, \ldots, N,$$

using the induced p-value

$$\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i\leftarrow\mathcal{O}(i)}\right) = \min\left\{\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i\leftarrow j}\right) : j \in \mathcal{O}(i)\right\}.$$

Analogously, the induced p-values for the TIX measures in (28) and the AIX measure in (29) are

$$\begin{aligned}
\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{\mathcal{O}(j)\leftarrow j}\right) &= \min\left\{\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i\leftarrow j}\right) : i \in \mathcal{O}(j)\right\}, \quad j = 1, \ldots, N, \\
\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{\mathcal{G}}\right) &= \min\left\{\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i\leftarrow j}\right) : (i,j) \in \mathcal{I}_K\right\}.
\end{aligned}$$

Since each rejection of an aggregate null in this subsection requires rejection of at least one constituent node-level inter-group null, the FWER control established for Algorithm 4.1 carries over to these induced tests.

4.3 Aggregate group connectedness

The inter-group node-level connectedness measures can be aggregated to obtain summary measures of connectedness between groups. For each ordered pair of distinct groups (g, h), define directional connectedness to group g from group h as

$$C^H_{\mathcal{G}_g \leftarrow \mathcal{G}_h} = \sum_{\substack{i \in \mathcal{G}_g \\ j \in \mathcal{G}_h}} C^H_{i \leftarrow j}, \quad g \neq h. \tag{30}$$

By cross-tabulating $C^H_{\mathcal{G}_g \leftarrow \mathcal{G}_h}$ for $g, h = 1, \dots, G$, $g \neq h$, we obtain a weighted directed network on the G aggregated nodes. Intra-group connectedness can also be reported descriptively, but it is not part of the family of hypotheses tested in this section.

Inference for aggregate group connectedness proceeds via induced p-values constructed from the adjusted node-level inter-group p-values $\tilde{p}_{\mathcal{P}}(\widehat{C}^H_{i \leftarrow j})$ for $(i, j) \in \mathcal{I}_K$. Specifically, for each ordered pair (g, h) with $g \neq h$, we test

$$\mathcal{H}_{\mathcal{G}_g \not\leftarrow \mathcal{G}_h} : C^H_{\mathcal{G}_g \leftarrow \mathcal{G}_h} = 0 \quad \text{versus} \quad \mathcal{H}_{\mathcal{G}_g \leftarrow \mathcal{G}_h} : C^H_{\mathcal{G}_g \leftarrow \mathcal{G}_h} > 0. \tag{31}$$

Because $C^H_{i \leftarrow j} \geq 0$, the null in (31) is equivalent to

$$C^H_{i \leftarrow j} = 0 \quad \text{for all } i \in \mathcal{G}_g, \ j \in \mathcal{G}_h.$$

We therefore define the induced p-value

$$\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{\mathcal{G}_g \leftarrow \mathcal{G}_h}\right) = \min\left\{\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i \leftarrow j}\right) : \ i \in \mathcal{G}_g, \ j \in \mathcal{G}_h\right\}, \quad g \neq h. \tag{32}$$

The group-to-group null in (31) is then rejected whenever at least one constituent node-level inter-group null is rejected.

By summing directional connectedness to group g over all source groups other than g, we obtain the group-level FIX measure

$$C^H_{\mathcal{G}_g \leftarrow \mathcal{G}_{-g}} = \sum_{\substack{s=1 \\ s \neq g}}^{G} C^H_{\mathcal{G}_g \leftarrow \mathcal{G}_s}, \tag{33}$$

where $\mathcal{G}_{-g}$ denotes aggregation over all groups except g. Similarly, the group-level TIX measure of total connectedness from group h to all other groups is

$$C^H_{\mathcal{G}_{-h} \leftarrow \mathcal{G}_h} = \sum_{\substack{r=1 \\ r \neq h}}^{G} C^H_{\mathcal{G}_r \leftarrow \mathcal{G}_h}, \tag{34}$$

and total inter-group connectedness at the group level is

$$C^H_{\mathcal{G}\leftarrow\mathcal{G}} = \sum_{\substack{r,s=1\\ r\neq s}}^{G} C^H_{\mathcal{G}_r\leftarrow\mathcal{G}_s}. \tag{35}$$

By definition of $C^H_{\mathcal{G}_r\leftarrow\mathcal{G}_s}$ in (30), we have

$$C^H_{\mathcal{G}\leftarrow\mathcal{G}} = \sum_{\substack{r,s=1\\ r\neq s}}^{G} \sum_{\substack{i\in\mathcal{G}_r\\ j\in\mathcal{G}_s}} C^H_{i\leftarrow j} = \sum_{(i,j)\in\mathcal{I}_K} C^H_{i\leftarrow j} = C^H_{\mathcal{G}},$$

where $C^H_{\mathcal{G}}$ is defined in (29).

The statistical significance of the measures in (33)–(35) is assessed using induced p-values constructed from the node-level adjusted p-values:

$$\begin{aligned}
\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{\mathcal{G}_g\leftarrow\mathcal{G}_{-g}}\right) &= \min\left\{\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i\leftarrow j}\right) : i\in\mathcal{G}_g,\ j\notin\mathcal{G}_g\right\},\\
\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{\mathcal{G}_{-h}\leftarrow\mathcal{G}_h}\right) &= \min\left\{\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i\leftarrow j}\right) : i\notin\mathcal{G}_h,\ j\in\mathcal{G}_h\right\},\\
\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{\mathcal{G}\leftarrow\mathcal{G}}\right) &= \min\left\{\tilde{p}_{\mathcal{P}}\left(\widehat{C}^H_{i\leftarrow j}\right) : (i,j)\in\mathcal{I}_K\right\}.
\end{aligned}$$

As before, each rejection of one of these aggregate null hypotheses implies rejection of at least one constituent node-level inter-group null. Hence the FWER control established for the underlying inter-group multiple-testing procedure carries over to the families of group-level FIX and TIX tests, as well as to the family of group-to-group tests in (31). The test of $C^H_{\mathcal{G}\leftarrow\mathcal{G}} = 0$ is therefore a level-α test.

5 Simulation experiments

In this section, we examine the empirical FWER and empirical average power achieved by the proposed multiple testing procedures. The average power is defined as the average expected number of correct rejections among all false null hypotheses (Bretz et al., 2011, Section 2.1). We consider VAR processes for $N = 20$ variables in model (1) with $\mathbf{c} = \mathbf{0}$, and vary the DGP configuration by letting $p = 1, 4$ and $T = 250, 500, 1000$. The connectedness measures are examined at horizons $H = 1, 10$.

We impose null hypotheses by choosing a number N_0 of nodes (or singleton groups) that are isolated from the rest of the network, $N_0 = 0, 5, 10, 15, 20$ in the simulation. These nodes belong to the set $\mathcal{N}_0 = \{i : C^H_{i\leftarrow j} = 0 \text{ and } C^H_{j\leftarrow i} = 0, \text{ for } j = 1,\ldots,N,\ j\neq i\}$.

The null hypotheses are imposed by setting to zero certain elements of the $\mathbf{\Phi}_\ell$ matrices and the $\mathbf{\Sigma}$ matrix. For each $i \in N_0$, we set $\phi_{ij,\ell} = \phi_{ji,\ell} = 0$, for $j = 1, \ldots, N$, such that $j \neq i$, and $\ell = 1, \ldots, p$. The non-zero elements of the $\mathbf{\Phi}_\ell$ matrices are generated randomly as $\phi_{ij,\ell} \sim \mathcal{U}(-1, 1)$. In order to obtain a covariance-stationary parametrization, the matrices $\mathbf{\Phi}_\ell$ are shrunk towards zero by halving their elements until all roots of the characteristic polynomial are outside the unit circle. The error covariance matrix is obtained by first generating a random vector $\mathbf{b} = (b_1, \ldots, b_N)'$ with elements $b_i \sim \mathcal{U}(-1, 1)$ and then setting $\mathbf{\Sigma} = \mathbf{I}_N + \mathbf{b}\mathbf{b}' - \text{diag}(\mathbf{b}\mathbf{b}')$. For each $i \in N_0$, we set $\sigma_{ij} = \sigma_{ji} = 0$, for $j = 1, \ldots, N$, such that $j \neq i$, to impose the null hypotheses. The positive definiteness of $\mathbf{\Sigma}$ is then ensured by shrinking it towards $\mathbf{I}_N$.[5]

We start the VAR recursions with zero values, simulate $1000 + p + T$ values of $\mathbf{y}_t$, and discard the first 1000 to mitigate the effects of the initial values. The simulation results are based on 1000 replications of each DGP configuration. The MC procedures use $B = 100$, and all the tests are conducted with $\alpha = 0.05$.

5.1 Null of isolated nodes

Table 1 reports the empirical rejection rates of the significance tests for node-level connectedness, which proceed by testing the $M = 380$ null hypotheses defined in (8). Panel A presents the results using the single-step method, and Panel B using the step-down adjustments. As expected from the developed theory, strong control of the FWER is maintained across different scenarios. The empirical FWER values are generally close to the nominal 5% level with few small exceedances. When all hypotheses under consideration are false ($N_0 = 0$), the FWER is by definition zero, as there are no true null hypotheses to reject incorrectly. As the number of isolated nodes N_0 increases, the empirical FWER gets closer to the nominal 5% for both the single-step and step-down methods.

The increased power with N_0 suggests that it becomes easier to detect non-zero connections when there are fewer truly non-zero connections in the underlying DGP. Moreover, the rejection rates under the alternative hypothesis of the single-step and step-down methods also converge, signaling that the power gain of the step-down method is more pronounced with lower values of N_0. It is important to note, however, that the concept of statistical power becomes moot when all hypotheses being tested are true ($N_0 = 20$), reflecting a scenario

[5]More specifically, we start with $\xi_0 = 1$ and $\mathbf{\Sigma}_0 = \mathbf{\Sigma}$, and then iterate on $\mathbf{\Sigma}_{k+1} = \xi_k \mathbf{\Sigma}_k + (1 - \xi_k)\mathbf{I}_N$, $\xi_{k+1} = \xi_k/2$ until the eigenvalues of $\mathbf{\Sigma}_{(k+1)}$ are all positive.

without any false null hypotheses to detect. Table 1 also shows that increasing p tends to reduce power due to the complexity added by considering more lagged terms in the VAR model, which dilutes the signal of connectedness. Conversely, increasing T and H improves the power, as larger samples provide more information, and longer horizons allow for more opportunities to detect true connectedness within the network. The simulation results also reveal that a sample size of $T = 250$ might be too small to achieve good power performance for the multiple testing procedure. For the most commonly used parameter setting in empirical applications, $p = 1$ and $H = 10$, we recommend a minimum sample size of $T = 500$.

Table 2 presents the empirical rejection rates of significance tests for the node-level FIX, TIX, and AIX measures, defined in (5)–(7). We only present results obtained using the step-down method, because the single-step method performs similarly with slightly lower power. The empirical FWERs are all near or below the nominal 5% threshold. As N_0 increases to 20, the empirical FWERs become identical for FIX, TIX, and AIX, which also coincide with the FWERs in Table 1. For the AIX, the rejection rates correspond to the empirical size and power of the test as there is a single null hypothesis, which can only be true when $N_0 = 20$.

The empirical average power of the tests for the FIX, TIX, and AIX measures varies quite markedly in Table 2. The TIX exhibits lower power compared to FIX in most scenarios, which shows that detecting outgoing connectedness is more challenging than incoming connectedness. Interestingly, the power for the TIX test increases consistently with N_0 across all values of p and H. In contrast, the FIX test tends to lose power as N_0 grows when $H = 1$, suggesting that incoming connectedness becomes relatively more challenging to detect in more fragmented networks at short forecast horizons. AIX stands out in Table 2 with its near 100% power in almost all scenarios, with a slight decrease observed only at $N_0 = 15$. This underscores AIX's comprehensive nature as a measure of overall network connectedness.

5.2 Null with groups

We now examine the performance of the testing procedures when a group of nodes is excluded from testing in the null hypothesis. As shown in Section 4, imposing groups in the network should improve the power of the tests, as there are fewer edges competing for significance. Strong control of FWER still holds. Therefore, we omit the empirical FWER in these simulations, and only report the average power in Tables 3–6.[6]

[6] For the aggregated indices shown in Tables 4 and 6, the results using the single-step adjustments are also omitted because they are similar to those of the step-down method.

We first investigate the power performance when testing pairwise connectedness at the node level. Table 3 reports the empirical average power of the significance tests for node connectedness when we designate a primary group of size $|\mathcal{G}_1| = 5$ and then 10, along with $20 - |\mathcal{G}_1|$ remaining singleton groups. These results are comparable to Table 1, which corresponds to $|\mathcal{G}_1| = 1$. Here we only test for the significance of the inter-group connectedness, i.e., the $K = 20^2 - |\mathcal{G}_1|^2 - (20 - |\mathcal{G}_1|)$ null hypotheses appearing in (21). The value of N_0 in Table 3 refers to the number of truly isolated nodes among the $20 - |\mathcal{G}_1|$ singleton groups. This means that power becomes irrelevant when $N_0 = 20 - |\mathcal{G}_1|$; i.e., when all the null hypotheses appearing in (21) are in fact true. Keeping the same N_0 value, results in Table 3 show that reducing K by increasing the size of the primary group to 10 nodes improves the power of the significance tests for node connectedness across all scenarios. This can also be observed by comparing with scenarios with smaller primary group sizes in Table 1 across the same value of N_0. Furthermore, Table 3 reveals that the step-down method consistently offers a slight, yet discernible, edge in terms of power over the single-step method.

Table 4 reports the power of step-down tests for node-level FIX and TIX in the presence of groups. Here the power for the AIX significance test is uniformly 100%, and hence is not displayed. The results show that increasing the size of the primary group from $|\mathcal{G}_1| = 5$ to $|\mathcal{G}_1| = 10$ tends to result in power losses for the step-down significance tests. This is particularly true for the FIX measure. The drop in power seen in Table 4 is due to the fact that (27) and (28) accumulate fewer connectedness measures when larger groups are designated.

We next examine the power performance of significance tests for group-level connectedness discussed in Section 4.3. Table 5 summarizes the average power of the significance tests for the group-level connectedness in the resulting networks with $G = 16$ and $G = 11$ groups when $|\mathcal{G}_1| = 5$ and $|\mathcal{G}_1| = 10$, respectively. In the context of (31), this analysis involves testing a total of $A = 240$ null hypotheses for the former and $A = 110$ null hypotheses for the latter scenario. Compared to Table 1, results in Table 5 indicate that increased size of the primary group $|\mathcal{G}_1|$ enhances the power of both the single-step and step-down methods, demonstrating that it is easier to detect connectedness at the group level than at the individual node level. This follows because there are fewer hypotheses under test, and the aggregate measures in (30) provide a stronger signal due to the cumulative effects of multiple connections between groups. As with the pattern observed in Table 1, increasing the lag length of the VAR model tends to reduce the power of the tests, while longer forecast horizons and larger sample sizes are associated with increased power.

Table 6 presents the average power of step-down significance tests for the group-level FIX and TIX measures in (33) and (34).[7] Comparing Tables 2 and 6 illuminates how the granularity of analysis impacts the detection of connectedness in a network. Notably, both FIX and TIX measures at the group level exhibit high power across the various scenarios. It is also evident that increasing the size of the primary group from 5 to 10 boosts the power of the tests for both measures. Consistent with our findings from Table 2, the FIX tends to show higher power than TIX, suggesting higher sensitivity in detecting incoming connectedness.

6 Empirical Applications

We undertake two applications of our step-down multiple testing procedure. First, we revisit the original equity return network constructed by Diebold and Yilmaz (2009) for a sample of 19 global stock markets from 1992 to 2007. Our method uncovers new details of the network topology that are obscured using point estimates alone, including an underlying group structure and the presence of a largely isolated node. Next, we use more recent data from 2003 to 2023 to analyze volatility connectedness among the G20 stock markets. Borrowing from the contagion testing literature (e.g., Forbes and Rigobon, 2002; Fry et al., 2010), we test for changes in the topology of the network between a benchmark (stable) period and a comparison (crisis) period. This allows us to test for the creation or removal of edges over time, which affects the channels through which shocks can spread among markets. This brings the DY framework for empirical network analysis closer to the theoretical literature that emphasizes the processes by which links are created and deleted and the aggregate implications of the structure of economic and financial networks.[8] In this way, we demonstrate marked topological change in both the GFC and COVID periods, as well as important differences in the structure of the volatility network between these two episodes.

6.1 Inference for the Diebold and Yilmaz (2009) network

We investigate the statistical significance of all pairwise connections among 19 global equity markets from January 10, 1992 to November 23, 2007 using the original data set from Diebold and Yilmaz (2009). As in their paper, we fit a VAR(1) to 829 weekly log-returns of the stock

[7] The power for the group-level AIX test was again found to be uniformly 100%, and is thus not displayed.

[8] See Glasserman and Young (2016) and Jackson and Pernoud (2021) for excellent surveys, as well as the notable contributions of Allen and Gale (2000), Gai and Kapadia (2010) and Acemoglu et al. (2015).

indices.[9] To conform with current practice, we replace the Cholesky factorization used by Diebold and Yilmaz (2009) with the GFEVD in (3), thereby achieving order-invariance.

In Table 7, we tabulate the connectedness matrix for the forecast horizon $H = 10$ weeks. Columns (rows) correspond to the originating (receiving) market of the spillover effect, while the pink/red shading convey information on the level of insignificance using our step-down procedure with $B = 1000$. Meanwhile, cells on the prime diagonal that are excluded from the test family are shaded light purple.

First, consider the group of advanced economy stock markets, including the US, the UK, France (FR), Germany (DE), Hong Kong (HK), Japan (JP), Australia (AU), and Singapore (SG). Shocks to any of these markets generate non-negligible spillovers onto other markets within the group. Except for Japan, they also generate strong spillovers onto many emerging markets. Three groups of emerging markets are apparent: (i) the Asian cluster, including Indonesia (ID), South Korea (KR), the Philippines (PH), Taiwan (TW), Thailand (TH), and Malaysia (MY); (ii) the Latin American cluster, including Argentina (AR), Brazil (BR), Chile (CL), and Mexico (MX); and (iii) Turkey (TR), which forms a group on its own. The two regional emerging market clusters are characterized by strong intra-regional spillovers but mostly insignificant inter-regional spillovers. The only exception is Mexico, which generates spillovers to both advanced and emerging markets. By contrast, the Turkish market generates no significant spillovers and its TIX is insignificant at the 5% FWER level.

Because we are testing null hypotheses of zero spillovers, we are able to introduce sparsity into the network. In Table 8, we exclude all edges that are insignificant at the 1% FWER level.[10] This necessitates a further renormalization of the spillover matrix to restore its grand sum to 100%. To establish concise terminology, we will refer to spillover statistics obtained prior to this renormalization step as *raw* spillovers, and those obtained after as *adjusted* spillovers. Due to this additional renormalization, if one or more edges are insignificant, then the adjusted spillovers will take larger values than their raw counterparts. By implicitly treating all edges as significant, as is the common practice in the DY literature, one will underestimate the magnitude of significant edges. Furthermore, the raw AIX will exceed the adjusted AIX if one or more bilateral spillovers are insignificant. Indeed, the raw AIX in Table 7 is about 4.5

[9] Diebold and Yilmaz (2009) also analyze volatility spillovers. Rather than replicating that exercise, we use an updated dataset to model volatility spillovers among global stock markets in our second empirical exercise, the results of which are presented in Subsection 6.2.

[10] Among the 342 off-diagonal elements in the connectedness table, 44 are insignificant at the 10% FWER level, 61 are insignificant at the 5% FWER level, and 129 are insignificant at the 1% FWER level. Therefore, our inferential procedure allows us to eliminate roughly 38% of pairwise connections as insignificant at the 1% FWER level.

ppts higher than the adjusted AIX at the 1% FWER level in Table 8. It follows that network inference is important even if one is only interested in aggregate spillover statistics.

We plot both the raw network and the adjusted network obtained at the 1% FWER level in Figure 1. The inherent density of the raw network shown in panel (a) makes it complex and difficult to interpret. This has led some authors to adopt arbitrary selection methods when generating network graphics, such as only plotting the strongest spillovers (e.g. Diebold and Yilmaz, 2014; Bostanci and Yilmaz, 2020). Our procedure improves on this type of arbitrary thresholding by providing a statistical method to distinguish significant spillover effects from noise. Figure 1(b) plots the adjusted network at the 1% FWER level from Table 8.

Figure 1 has several features to assist the reader. Each node is colored blue or red. Blue (red) shading denotes markets with a positive (negative) net spillover effect. Node size is proportional to the total connectedness of each market (TIX+FIX). We color-code the edges as follows: (i) blue for edges originating in the group of advanced economy markets; (ii) yellow for edges originating in the Asian group; (iii) green for edges originating in the Latin-American group; and (iv) purple for edges originating in Turkey. The thickness of the lines and the size of the arrows are proportional to the strength of the directional spillover effect.

The figure shows that within-group edges are typically stronger than inter-group edges. We now exploit this observation by imposing the group structure on the network and re-testing at both the node and group levels. In Table 9, we report the adjusted node-level pairwise connectedness estimates at the 1% FWER level.[11] Cells shaded pale purple indicate intra-group connections exempt from testing. Comparing Tables 8 and 9 reveals an important phenomenon. We find more statistically significant inter-group relationships once the group structure is imposed. This is expected, as the group structure reduces the number of off-diagonal elements competing for significance, thereby improving the power of the test.

Table 9 also shows that intra-group connectedness is particularly strong among the advanced markets. The TIX measures for most of these markets are reduced by more than half when only the inter-group spillovers are considered. At the aggregate systemwide level, almost half of spillover activity is due to intra-group links. The adjusted AIX drops from 61.08% when considering all pairwise edges to 32.19% using only inter-group edges.

[11]Note that the point estimates of the pairwise spillovers in Table 9 differ from those reported in Tables 7 and 8 due to the exclusion of insignificant edges conditional on the group structure and subsequent renormalization. Note also that the aggregate spillover indices reported in this table capture the connectedness between nodes that are in different groups. See Appendix Table B.1 for a version of this table where insignificant edges are not excluded. The point estimates of the pairwise spillovers in that case are identical to those in Table 7 by construction.

In Table 10, we report adjusted group-level connectedness at the 1% FWER level. The dominance of the advanced markets in the global return network is easily seen. Spillovers among the advanced markets account for 30.77% of systemwide connectedness and their outward spillovers a further 17.38%. Spillovers within the groups of Asian and Latin American markets are also relatively large (19.15% and 11.68%, respectively), but the outward spillovers they generate are much weaker. The isolation of Turkey is again evident.

This exercise shows that our testing framework can reveal previously unknown topological details in Diebold and Yilmaz's seminal 2009 study of equity market connectedness, including an underlying group structure and the presence of a largely disconnected market. Next, we will show how our approach can be used to test for topological change over time.

6.2 Link creation and deletion in the G20 equity volatility network

Connections between nodes in financial networks can either promote risk-sharing or act as channels of contagion, depending on the features of the network (e.g., Allen and Gale, 2000; Gai and Kapadia, 2010; Acemoglu et al., 2015). Consequently, the creation and/or deletion of links between nodes alters the channels available for risk-sharing and shock propagation and may change the relative importance of nodes in the network. The ability to detect such changes in DY networks may be of use to investors making portfolio allocation decisions and to policy makers and regulators charged with maintaining financial stability.

We continue with the study of equity market connectedness using a daily data set covering the G20 stock markets from July 25, 2003 to July 28, 2023. The G20 is an international forum that encompasses the world's largest economies, representing approximately 85% of global GDP, 75% of international trade, and two-thirds of the world's population.[12] There are 19 sovereign states in the G20, as well as the European Union and the African Union. Our sample includes stock market data for all of the sovereign states except Saudi Arabia, which uses the Islamic calendar with regular weekend trading. We exclude the European Union to avoid double-counting with respect to the three European sovereigns in the G20, and we exclude the African Union due to the absence of a suitable unified stock index. Therefore, our sample contains 18 markets, as detailed in Table C.1 of Appendix C.

Following Diebold and Yilmaz (2009) and Demirer et al. (2018), we calculate the range-

[12] For further details, see https://www.dfat.gov.au/trade/organisations/g20.

based realized volatility of each market as follows:

$$
\begin{aligned}
V_t =& 0.511(H_t - L_t)^2 - 0.383(C_t - O_t)^2 \\
& - 0.019\Big((C_t - O_t)(H_t + L_t - 2O_t) - 2(H_t - O_t)(L_t - O_t)\Big),
\end{aligned} \tag{36}
$$

where O_t, C_t, H_t and L_t are the logs of the daily opening, closing, high, and low prices on day t, respectively. Our sample period includes two crisis episodes in which we can test for evidence of topological change: (i) the Global Financial Crisis (GFC) and; (ii) the COVID pandemic. To implement our test, we follow the approach of the contagion testing literature and test for statistically significant changes in inter-market linkages between a benchmark pre-crisis period and a crisis period. Notable examples of this approach include King and Wadhwani (1990), Forbes and Rigobon (2002), Bekaert et al. (2005) and Fry et al. (2010).

Our subsample selection is as follows:

1. The GFC:

   (a) the pre-crisis period covers the 1,034 trading days starting at the beginning of our sample on July 25, 2003 and ending on July 25, 2007. This is a period of relative stability that starts with the economic recovery after the dotcom bubble and the Enron and WorldCom scandals and ends just before Bear Sterns liquidated two of its hedge funds due to their exposure to collateralized debt obligations.[13]

   (b) the crisis period covers the 627 trading days from July 26, 2007 to December 31, 2009, which encompasses the key events of the crisis and much of the slow recovery documented by Burns et al. (2010) but ends before concerns about European sovereign debt sustainability began to escalate in 2010.

2. The COVID pandemic:

   (a) the pre-crisis period covers the 770 trading days from January 2, 2017 (which excludes the volatile period following the Brexit referendum in June 2016) until December 30, 2019, just before the WHO became aware of cases of 'viral pneumonia' in Wuhan.[14]

   (b) the crisis period covers the 865 trading days from December 31, 2019 to May 5,

[13]See Bear Stearns Seizes Troubled Hedge Fund's Assets.

[14]See Timeline: WHO's COVID-19 response.

2023, when the WHO declared that the COVID pandemic no longer represented a global health emergency.[15]

A VAR model of order $p = 1$ is fitted to the realized volatilities in each subsample and the connectedness measures are computed for $H = 10$ days. The lag length $p = 1$ is chosen using the BIC and HQ information criteria. First, consider the GFC. In panels (a) and (b) of Figure 2, we plot the adjusted bilateral connections that are significant at the 1% FWER level in the pre-crisis and crisis sample periods. As before, the size of the nodes is proportional to the total connectedness of each market (TIX+FIX), with blue (red) shading denoting positive (negative) net spillover. The directional connectedness measures are shown in four different colors: (i) blue for connections originating from the group of North American and European advanced economies (NA-EU); (ii) yellow from the Asia-Pacific markets, including Japan, South Korea, and Australia (APAC); (iii) green from the Latin American markets (LATAM); and (iv) pink from Russia, South Africa, and Turkey (OTHER). Meanwhile, in panel (c), we present a simple visualization of the change in statistical significance of each element of the connectedness table from the pre-crisis period to the crisis period. To conserve space, the full spillover tables for each subsample are placed in Appendix C.

Panel (a) of Figure 2 reveals strong bilateral volatility spillovers among the US, UK, France (FR), Germany (DE) and Italy (IT) prior to the GFC. At the other end of the spectrum, China (CN) stands isolated from the other 17 markets, neither generating nor receiving any statistically significant spillover effects. The finding that a group of major stock indices form a group that accounts for a large proportion of cross-market spillover while smaller markets and emerging markets typically generate weak spillover effects is well-documented in the DY literature (see chapter 4 of Diebold and Yilmaz, 2015, for a notable example). The adjusted AIX at the 1% FWER level reveals that 30.47% of the total network connectedness is due to bilateral spillovers in the pre-GFC period, with more than two-thirds of the systemwide 10-days-ahead forecast error variance being attributed to each markets' own shocks.

The comparison against the GFC period is stark. The adjusted AIX at the 1% FWER level rises from 30.47% to 50.49%, indicating a marked increase in the relative intensity of volatility spillovers. This intensification of cross-market shock propagation is consistent with the large body of literature that documents increased asset comovement in adverse states of the world (e.g. Longin and Solnik, 2001; Ang et al., 2006). It is also a common finding

[15]See WHO Director-General's opening remarks at the media briefing, 5 May 2023.

in the DY literature, with the observation of volatility bursts among equity markets being a prominent feature in the foundational work of Diebold and Yilmaz (2009). However, Figure 2 reveals that the increase in volatility spillovers during the GFC is not broad-based; rather, it is concentrated among the NA-EU group, while spillovers originating from the Asia-Pacific region and the emerging markets are greatly diminished.

To better appreciate the change in network topology, in Figure 2(c) we present a *significance-transition map*: a matrix in which each cell indicates whether the corresponding directed edge has gained significance (pink), lost significance (grey), remained significant (white), or remained insignificant (marked with an 'X') between the pre-crisis and GFC periods. This representation provides a compact and transparent visualization of topological change that complements the network graphs in panels (a) and (b).

It is striking that the large majority of newly created links originate in the NA-EU markets, with many new links to the emerging markets being formed. The creation of additional links originating in the LATAM markets also leads to their increasing integration with NA-EU markets. By contrast, the large majority of deleted links originate in India, Turkey, South Africa and Russia, none of which generate any significant bilateral spillovers at the 1% FWER level during the GFC. In fact, we find that the bilateral spillovers between the Japanese and Korean markets—both important regional benchmarks—are the only edges originating in the Asia-Pacific to remain statistically significant at the 1% FWER level during the GFC.

To place these changes in context, we report a range of summary statistics in Table 11. Prior to the GFC, 25.5% of all possible edges (78/306) are statistically significant at the 1% FWER level. This drops to 24.2% (74/306) during the GFC. In total, 35 edges are deleted and 31 created. Link deletion occurs in every region, with the removed edges collectively accounting for 9.29ppts of system-wide forecast error variance. By contrast, link creation is confined to the NA-EU and LATAM markets (23 and 8 new links, respectively). Newly created links originating in the former (latter) account for 14.87 (4.54) ppts of system-wide forecast error variance. Consequently, the change in aggregate spillover intensity in the GFC is driven by the removal of a large number of weak links and the creation of a smaller number of much stronger links. This explains why the adjusted AIX at the 1% FWER level increases from 30.47% to 50.49% despite the increased sparsity of the network.

These topological changes reflect the origin of the GFC in the US subprime mortgage market and the considerable exposure of many developed market financial institutions to the distressed US securitized asset markets (Brunnermeier, 2009). Furthermore, the emergence of fewer but stronger links and the greater centrality of the NA-EU markets alters the channels of volatility transmission, creating conditions that may amplify contagion from these markets.

Moving on to the COVID pandemic, Figures 3(a) and (b) show the adjusted bilateral spillovers that are significant at the 1% FWER level in the pre-COVID and COVID sample periods. The network structure in the pre-COVID period is similar to that seen in the pre-GFC period (Figure 2(a)) in the sense that most spillover activity occurs among the NA-EU markets, with relatively few significant spillovers originating elsewhere. The network is slightly more sparse, with 19.0% (58/306) of all possible edges being significant at the 1% FWER level (see Table 11). The adjusted AIX at the 1% FWER level is also slightly lower, at 26.89% pre-COVID compared to 30.47% pre-GFC.

During the COVID crisis, the network becomes much denser, a change that is clearly visible in Figure 3(b). At this time, 32.4% (99/306) of possible edges are significant at the 1% FWER level and the adjusted AIX almost doubles, from 26.89% to 48.73%. Figure 3(c) and Table 11 reveal that the majority of new links originate in the NA-EU markets (31 new links). This was also true of the GFC. However, there are two major differences between the COVID crisis and the GFC: link creation is also observed from the APAC markets during the COVID period (10 new links), and far fewer links are deleted than during the GFC (14 compared to 35). As a result, link deletion removes just 4.47ppts of spillover activity during the COVID crisis, while link creation adds 20.39ppts.

This highlights an interesting distinction between the GFC and the COVID crisis. During the GFC, many weak links were removed but spillover intensity increased nonetheless due to the presence of a relatively small number of strong links. By contrast, the increase in spillover intensity during the COVID pandemic results from the proliferation of weaker spillovers, leading to an increase in the density of the network. This denser structure provides a greater number of direct avenues for volatility transmission, allowing for rapid shock propagation even though each link in the network is weaker on average. This is an interesting finding that may reflect the different origins of the two crises. While the GFC was a financial crisis that directly impacted the financial markets through which it spread, the COVID pandemic was first and foremost a public health emergency with substantial real economic effects (both realized and anticipated) that were subsequently amplified by the financial markets (Ramelli and Wagner, 2020). Furthermore, the unprecedented and sweeping government restrictions on economic and social activity enacted around the world that Baker et al. (2020) identify as a key contributor to large stock market losses during the COVID pandemic may contribute to the broad-based increase in spillover activity that we observe.

7 Conclusion

The DY framework for network analysis represents a major contribution to the financial economist's toolkit. In this paper, we set out to enrich their method with the addition of a robust inferential framework that accounts for the multiplicity of inferences inherent in testing hypotheses regarding network connectedness. We develop both single-step and step-down multiple testing procedures in which MC resampling is used to obtain exact p-values with strong control of the FWER. Our approach enables the identification of underlying sparsity within the network and can accommodate arbitrary group structures but does not assume either.

We conduct two empirical analyses that highlight different aspects of the utility of our approach. First, we revisit Diebold and Yilmaz's foundational 2009 analysis of global equity market return spillovers in search of additional structure in the estimated network that is invisible in the absence of network inference. We find that a large proportion of the estimated bilateral spillovers are statistically insignificant at standard levels, revealing previously unknown sparsity in the network. Furthermore, we uncover an underlying group structure in the data whereby markets are clustered according to their location and advanced/emerging market status. Finally, we show that the Turkish market is a largely isolated node that receives minimal spillover from other markets and generates none in its own right. Collectively, these previously hidden features of the network materially affect its properties.

In our second application, we test for link creation and deletion in the global equity volatility network between benchmark (stable) and comparison (crisis) sample periods, an approach that builds on the contagion testing literature (e.g., Forbes and Rigobon, 2002; Fry et al., 2010). In this way, we show that many weak links were deleted from the volatility network during the GFC, while fewer stronger links were created, most of which originated from the advanced economy stock markets. By contrast, the COVID period was characterized by the proliferation of many comparatively weak links. In both cases, aggregate spillover activity increases, but the pattern of volatility transmission differs substantially.

By introducing a method to separate link existence from link strength in the DY framework, our work lays the foundations for further contributions, both methodological and empirical. We particularly look forward to future work that expands our framework to accommodate high-dimensional VARs and that integrates insights from the theoretical literature on network formation into DY network models.

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Table 1: Rejection rates of the significance tests for node connectedness in a 20-node network with N_0 isolated nodes.

	$p=1, H=1$			$p=1, H=10$			$p=4, H=1$			$p=4, H=10$		
T	250	500	1000	250	500	1000	250	500	1000	250	500	1000
Panel A: Single-step method												
FWER												
$N_0 = 5$	1.6	2.0	1.4	0.6	1.0	0.9	1.5	2.0	1.7	0.3	0.4	0.6
10	3.7	3.5	2.9	1.3	1.1	1.1	2.1	3.9	2.8	0.7	0.8	1.2
15	4.7	4.0	5.6	3.6	2.0	3.7	3.9	4.1	4.8	1.6	1.8	3.0
20	5.8	5.1	6.2	4.6	4.4	4.8	5.7	5.3	4.1	4.2	4.7	5.5
Average power												
$N_0 = 0$	21.6	38.0	51.9	19.7	46.0	67.8	14.2	35.3	50.8	5.7	33.0	58.6
5	26.7	42.1	55.0	27.0	51.8	72.5	19.4	39.2	53.5	12.4	40.1	64.9
10	31.9	45.9	57.6	42.4	68.0	84.9	25.6	43.9	57.1	21.5	52.8	77.5
15	38.1	50.3	60.5	63.0	82.6	92.9	32.6	48.7	61.2	44.7	80.8	96.4
Panel B: Step-down method												
FWER												
$N_0 = 5$	1.8	2.3	1.8	0.7	1.1	1.3	1.5	2.8	2.5	0.3	0.4	1.0
10	4.0	3.7	3.4	1.6	1.4	1.7	2.4	4.3	3.0	0.9	1.0	1.5
15	4.7	4.1	5.6	3.7	2.2	4.0	4.0	4.1	5.2	1.6	1.8	3.0
20	5.8	5.1	6.2	4.6	4.4	4.8	5.7	5.3	4.1	4.2	4.7	5.5
Average power												
$N_0 = 0$	22.3	39.3	53.5	21.1	49.3	72.2	14.6	36.4	52.3	5.9	35.1	62.2
5	27.1	42.8	55.8	28.1	53.6	74.6	19.7	39.8	54.3	12.8	41.2	66.7
10	32.1	46.1	57.8	43.9	69.9	86.2	25.8	44.2	57.4	22.0	54.0	78.6
15	38.1	50.4	60.5	63.4	83.0	93.1	32.7	48.7	61.2	45.1	81.1	96.5

Notes: This table reports the empirical FWER and empirical average power (in percentages) of the significance tests for node connectedness ($C^H_{i\leftarrow j} > 0$) for 20-dimensional VAR(p) models, given a nominal FWER of 5%. There are $M = 380$ null hypotheses under test. The FWER is zero when there are no true null hypotheses to reject ($N_0 = 0$), indicating no risk of Type I errors. Conversely, power is irrelevant when all hypotheses are true ($N_0 = 20$), as there are no false null hypotheses to detect.

Table 2: Rejection rates of step-down significance tests for the node-level FIX, TIX, and AIX measures in a 20-node network with N_0 isolated nodes.

	$p=1, H=1$			$p=1, H=10$			$p=4, H=1$			$p=4, H=10$		
T	250	500	1000	250	500	1000	250	500	1000	250	500	1000
Panel A: FIX												
FWER												
$N_0 = 5$	1.8	2.3	1.8	0.7	1.1	1.3	1.4	2.8	2.5	0.3	0.4	1.0
10	4.0	3.7	3.3	1.6	1.4	1.6	2.4	4.2	2.9	0.9	1.0	1.4
15	4.6	4.0	5.6	3.6	1.9	3.7	4.0	4.1	5.2	1.4	1.8	2.9
20	5.8	5.1	6.2	4.6	4.4	4.8	5.7	5.3	4.1	4.2	4.7	5.5
Average power												
$N_0 = 0$	73.2	83.1	88.8	88.2	99.9	100	65.4	82.0	88.3	49.7	98.3	100
5	73.9	83.0	88.2	88.6	99.6	100	67.7	81.7	87.9	65.5	97.4	100
10	73.6	81.7	87.3	94.0	99.3	100	68.5	81.5	87.7	74.6	96.9	100
15	68.4	78.1	84.4	96.9	99.6	100	64.0	77.8	85.4	85.8	99.4	100
Panel B: TIX												
FWER												
$N_0 = 5$	0.8	0.6	0.5	0.2	0.2	0.6	0.7	0.7	0.8	0.2	0.3	0.6
10	2.5	2.2	1.2	1.1	1.2	0.9	1.4	2.5	1.3	0.7	0.8	0.9
15	4.4	3.3	4.4	2.5	2.0	3.2	3.1	3.0	3.0	1.6	1.7	2.4
20	5.8	5.1	6.2	4.6	4.4	4.8	5.7	5.3	4.1	4.2	4.7	5.5
Average power												
$N_0 = 0$	44.9	62.5	74.0	46.1	82.8	98.4	36.2	60.0	72.8	20.0	56.0	87.8
5	52.4	66.5	76.9	56.0	86.5	98.5	44.4	64.5	75.7	33.6	64.6	91.6
10	59.3	71.1	79.7	73.4	93.3	99.2	53.2	69.8	79.5	47.4	79.3	95.9
15	63.6	74.6	82.0	89.8	98.1	99.7	59.3	73.3	82.4	73.2	96.7	99.9
Panel C: AIX												
Size												
$N_0 = 20$	5.8	5.1	6.2	4.6	4.4	4.8	5.7	5.3	4.1	4.2	4.7	5.5
Power												
$N_0 = 0$	100	100	100	100	100	100	100	100	100	97.9	100	100
5	100	100	100	99.6	100	100	100	100	100	99.7	100	100
10	100	100	100	100	100	100	99.9	100	100	99.3	100	100
15	93.3	97.1	98.7	100	100	100	91.9	97.0	99.4	99.7	100	100

Notes: This table reports the empirical FWER and empirical average power (in percentages) of step-down significance tests for the node-level FIX, TIX, and AIX measures for 20-dimensional VAR(p) models, given a nominal FWER of 5%. For the AIX, the rejection rates correspond to the empirical size and power. The FWER is zero when there are no true null hypotheses to reject ($N_0 = 0$), indicating no risk of Type I errors. Conversely, power is irrelevant when all hypotheses are true ($N_0 = 20$), as there are no false null hypotheses to detect.

Table 3: Average power of significance tests for node connectedness in a 20-node network comprising a primary group of size $|\mathcal{G}_1|$ and $20 - |\mathcal{G}_1|$ singleton groups, and with N_0 isolated singleton groups.

	$p=1, H=1$			$p=1, H=10$			$p=4, H=1$			$p=4, H=10$		
T	250	500	1000	250	500	1000	250	500	1000	250	500	1000
Panel A: $\lvert\mathcal{G}_1\rvert = 5$												
Single-step method												
$N_0 = 0$	22.2	38.5	52.1	21.5	47.5	68.9	14.7	35.5	51.0	6.9	34.7	60.0
5	27.0	42.4	55.1	29.1	54.0	73.5	19.9	39.3	53.4	13.9	41.4	66.0
10	32.4	46.3	57.9	46.4	70.9	86.6	25.5	43.0	55.7	23.8	54.9	78.7
Step-down method												
$N_0 = 0$	22.9	39.9	54.0	23.1	50.9	73.4	15.2	36.9	52.8	7.2	36.9	63.7
5	27.5	43.1	55.8	30.0	55.1	75.1	20.2	40.0	54.0	14.2	42.6	68.0
10	32.6	46.5	58.1	47.4	72.2	87.2	25.6	43.2	55.9	24.2	55.6	79.2
Panel B: $\lvert\mathcal{G}_1\rvert = 10$												
Single-step method												
$N_0 = 0$	23.5	39.3	52.8	25.2	50.3	70.9	16.2	36.8	52.0	9.5	38.0	62.5
5	28.5	43.4	55.7	32.9	56.3	75.4	21.8	40.5	54.4	17.4	44.9	68.6
Step-down method												
$N_0 = 0$	24.3	40.8	54.7	26.9	53.9	75.9	16.6	38.2	53.9	9.9	40.5	66.5
5	28.9	43.9	56.3	33.6	57.5	76.7	22.0	41.0	54.9	17.6	45.7	69.7

Notes: This table reports the empirical average power (in percentage) of the of the significance tests for node connectedness for 20-dimensional VAR(p) models, given a nominal FWER of 5% when accounting for the presence of groups. This entails the testing of $K = 360$ and $K = 290$ null hypotheses when $|\mathcal{G}_1| = 5$ and $|\mathcal{G}_1| = 10$, respectively. Power is irrelevant in Panel A when $N_0 = 15$ and in Panel B when $N_0 = 10$, as there are no false null hypotheses to detect in those cases.

Table 4: Average power of step-down significance tests for the node-level FIX, TIX, and AIX measures in a 20-node network comprising a group of size $|\mathcal{G}_1|$ and $20 - |\mathcal{G}_1|$ singleton groups, and with N_0 isolated singleton groups.

	$p=1, H=1$			$p=1, H=10$			$p=4, H=1$			$p=4, H=10$		
T	250	500	1000	250	500	1000	250	500	1000	250	500	1000
Panel A: $\|\mathcal{G}_1\| = 5$												
FIX												
$N_0 = 0$	73.4	83.3	88.7	89.8	99.9	100	65.5	82.1	88.4	55.2	98.6	100
5	73.4	82.7	88.2	89.4	99.4	100	67.3	80.8	87.2	67.5	97.4	100
10	71.1	80.8	86.9	93.1	98.9	100	65.5	78.7	85.6	72.3	96.1	99.9
TIX												
$N_0 = 0$	45.3	62.9	74.2	48.4	84.0	98.5	37.0	60.6	73.2	22.2	57.8	88.9
5	52.0	66.8	76.6	57.3	86.4	98.2	44.5	64.4	75.2	35.1	65.8	91.5
10	58.2	70.7	79.0	74.3	93.1	98.7	51.3	68.4	77.7	48.6	78.9	95.1
Panel B: $\|\mathcal{G}_1\| = 10$												
FIX												
$N_0 = 0$	73.1	83.2	88.8	91.4	99.8	100	65.5	81.9	88.4	61.1	98.9	100
5	69.4	80.6	86.9	85.1	98.2	99.9	61.9	78.5	85.8	62.4	95.1	99.8
TIX												
$N_0 = 0$	46.3	63.3	74.6	51.7	84.7	98.3	38.0	60.9	73.6	25.7	60.6	89.7
5	51.3	65.9	75.9	56.8	82.6	96.1	43.9	63.1	74.4	35.9	64.9	88.8

Notes: This table reports the empirical average power (in percentage) of step-down significance tests for the node-level FIX and TIX measures for 20-dimensional VAR(p) models, given a nominal FWER of 5% when accounting for the presence of groups. Power is irrelevant in Panel A when $N_0 = 15$ and in Panel B when $N_0 = 10$, as there are no false null hypotheses to detect in those cases. The power for the AIX test is uniformly 100%, and is thus not displayed.

Table 5: Average power of significance tests for aggregate group connectedness in a 20-node network comprising a group of size $|\mathcal{G}_1|$ and $20 - |\mathcal{G}_1|$ singleton groups, and with N_0 isolated singleton groups.

	$p = 1, H = 1$			$p = 1, H = 10$			$p = 4, H = 1$			$p = 4, H = 10$		
T	250	500	1000	250	500	1000	250	500	1000	250	500	1000
Panel A: $\|\mathcal{G}_1\| = 5$												
Single-step method												
$N_0 = 0$	25.8	42.4	55.7	25.3	52.0	72.3	17.6	39.3	54.4	8.8	38.6	63.4
5	32.8	48.2	60.2	35.4	59.9	78.1	24.9	44.8	58.2	18.3	47.3	70.6
10	43.2	56.6	67.0	58.2	79.3	90.9	35.1	52.7	64.0	34.1	64.7	84.4
Step-down method												
$N_0 = 0$	26.6	43.8	57.4	27.0	55.3	76.5	18.1	40.6	56.1	9.1	40.8	66.9
5	33.2	48.9	60.9	36.4	61.4	79.4	25.2	45.4	58.9	18.6	48.4	72.0
10	43.4	56.9	67.2	59.1	80.1	91.4	35.3	52.9	64.2	34.5	65.3	84.8
Panel B: $\|\mathcal{G}_1\| = 10$												
Single-step method												
$N_0 = 0$	29.8	45.3	57.8	32.9	57.3	75.9	22.4	43.1	57.3	14.9	45.3	68.1
5	40.4	53.9	64.8	47.4	68.6	83.5	34.1	51.8	63.6	30.1	57.7	77.9
Step-down method												
$N_0 = 0$	30.6	46.7	59.7	34.7	60.6	80.1	22.9	44.5	59.1	15.4	47.6	71.7
5	40.7	54.4	65.3	48.0	69.5	84.3	34.4	52.1	64.1	30.5	58.6	78.7

Notes: This table reports the empirical average power (in percentage) of the significance tests for group-level connectedness for 20-dimensional VAR(p) models, given a nominal FWER of 5%. The resulting networks comprise $G = 16$ and $G = 11$ groups when $|\mathcal{G}_1| = 5$ and $|\mathcal{G}_1| = 10$, respectively. This entails testing $A = 240$ null hypotheses for the former and $A = 110$ null hypotheses for the latter scenario. Power is irrelevant in Panel A when $N_0 = 15$ and in Panel B when $N_0 = 10$, as there are no false null hypotheses to detect in those cases.

Table 6: Average power of step-down significance tests for group-level FIX, TIX, and AIX measures in a 20-node network comprising a group of size $|\mathcal{G}_1|$ and $20 - |\mathcal{G}_1|$ singleton groups, and with N_0 isolated singleton groups.

	$p=1, H=1$			$p=1, H=10$			$p=4, H=1$			$p=4, H=10$		
T	250	500	1000	250	500	1000	250	500	1000	250	500	1000
Panel A: $\lvert\mathcal{G}_1\rvert = 5$												
FIX												
$N_0 = 0$	75.4	84.6	89.5	91.0	99.9	100	67.9	83.7	89.5	58.7	98.8	100
5	76.5	84.7	89.5	91.3	99.6	100	70.8	83.1	88.8	71.6	97.9	100
10	78.5	85.7	90.3	95.8	99.6	100	73.4	83.5	89.0	80.3	97.6	100
TIX												
$N_0 = 0$	49.0	65.7	76.2	52.3	85.9	98.8	40.4	63.1	75.2	25.5	60.6	90.1
5	57.3	70.6	79.3	62.6	89.0	98.8	49.3	67.9	77.8	40.4	69.9	93.2
10	67.3	77.1	83.7	81.0	96.0	99.6	59.8	74.5	82.3	58.0	84.4	96.7
Panel B: $\lvert\mathcal{G}_1\rvert = 10$												
FIX												
$N_0 = 0$	76.9	85.5	90.2	94.2	100	100	71.2	84.9	90.4	70.2	99.5	100
5	78.7	86.1	90.5	93.4	99.9	100	74.7	84.8	89.9	78.6	99.0	100
TIX												
$N_0 = 0$	52.4	67.4	77.6	58.9	89.2	99.2	44.7	65.7	77.0	33.3	66.8	93.2
5	62.4	73.4	81.5	69.8	91.5	99.4	56.7	71.7	80.3	50.3	76.0	95.4

Notes: This table reports the empirical average power (in percentage) of step-down significance tests for the group-level FIX and TIX measures for 20-dimensional VAR(p) models, given a nominal FWER of 5%. The resulting networks comprise $G = 16$ and $G = 11$ groups when $|\mathcal{G}_1| = 5$ and $|\mathcal{G}_1| = 10$, respectively. Power is irrelevant in Panel A when $N_0 = 15$ and in Panel B when $N_0 = 10$, as there are no false null hypotheses to detect in those cases. The power for the group-level AIX test is uniformly 100%, and is thus not displayed.

Table 7: Global stock market return connectedness.

	From																			FIX
To	US	UK	FR	DE	HK	JP	AU	SG	ID	KR	PH	TW	TH	MY	AR	BR	CL	MX	TR	
US	1.34	0.56	0.53	0.56	0.21	0.15	0.32	0.23	0.07	0.11	0.12	0.06	0.07	0.06	0.16	0.18	0.17	0.32	0.04	3.92
UK	0.52	1.24	0.73	0.66	0.29	0.15	0.30	0.22	0.05	0.11	0.10	0.06	0.09	0.06	0.14	0.12	0.12	0.24	0.06	4.02
FR	0.49	0.72	1.23	0.83	0.23	0.18	0.26	0.18	0.06	0.12	0.08	0.07	0.08	0.05	0.16	0.12	0.13	0.24	0.05	4.03
DE	0.51	0.65	0.83	1.22	0.26	0.15	0.25	0.21	0.07	0.12	0.09	0.09	0.09	0.07	0.12	0.13	0.11	0.23	0.05	4.04
HK	0.24	0.33	0.27	0.30	1.38	0.13	0.32	0.54	0.17	0.21	0.21	0.15	0.22	0.21	0.10	0.10	0.13	0.21	0.02	3.88
JP	0.25	0.28	0.31	0.28	0.20	2.08	0.31	0.24	0.13	0.25	0.06	0.16	0.07	0.11	0.10	0.16	0.08	0.16	0.04	3.18
AU	0.39	0.38	0.35	0.34	0.33	0.22	1.45	0.23	0.09	0.18	0.15	0.09	0.13	0.12	0.16	0.17	0.17	0.27	0.06	3.82
SG	0.24	0.26	0.21	0.24	0.51	0.15	0.22	1.31	0.24	0.23	0.31	0.17	0.33	0.32	0.13	0.07	0.12	0.17	0.03	3.96
ID	0.11	0.09	0.10	0.13	0.25	0.13	0.15	0.37	2.02	0.20	0.38	0.08	0.40	0.32	0.09	0.11	0.18	0.12	0.03	3.24
KR	0.17	0.19	0.20	0.20	0.30	0.23	0.24	0.34	0.17	1.98	0.10	0.19	0.33	0.11	0.10	0.09	0.10	0.15	0.06	3.29
PH	0.18	0.16	0.13	0.14	0.32	0.06	0.20	0.43	0.35	0.10	1.80	0.14	0.37	0.27	0.15	0.10	0.13	0.21	0.04	3.47
TW	0.15	0.14	0.18	0.23	0.29	0.18	0.13	0.33	0.09	0.23	0.17	2.36	0.15	0.16	0.10	0.05	0.10	0.17	0.05	2.90
TH	0.11	0.15	0.12	0.14	0.28	0.06	0.16	0.44	0.34	0.29	0.34	0.12	1.84	0.30	0.11	0.11	0.13	0.16	0.06	3.42
MY	0.10	0.11	0.09	0.14	0.33	0.10	0.17	0.50	0.36	0.12	0.29	0.13	0.37	2.05	0.09	0.03	0.12	0.16	0.01	3.21
AR	0.25	0.23	0.26	0.20	0.14	0.09	0.20	0.20	0.08	0.10	0.12	0.06	0.10	0.08	2.03	0.38	0.24	0.44	0.05	3.24
BR	0.28	0.21	0.21	0.23	0.14	0.16	0.25	0.11	0.09	0.10	0.09	0.04	0.12	0.03	0.39	2.07	0.33	0.35	0.07	3.20
CL	0.25	0.19	0.20	0.18	0.18	0.07	0.22	0.20	0.17	0.10	0.12	0.07	0.16	0.10	0.27	0.33	2.04	0.32	0.08	3.22
MX	0.37	0.31	0.31	0.31	0.24	0.11	0.27	0.21	0.08	0.12	0.16	0.09	0.12	0.12	0.33	0.27	0.20	1.59	0.03	3.67
TR	0.10	0.19	0.14	0.17	0.06	0.07	0.14	0.07	0.04	0.11	0.08	0.07	0.11	0.01	0.09	0.15	0.13	0.07	3.47	1.79
TIX	4.71	5.17	5.17	5.28	4.57	2.41	4.13	5.07	2.63	2.80	2.96	1.84	3.32	2.49	2.79	2.65	2.70	4.00	0.82	65.50

Notes: Sample period 10/01/1992 – 23/11/2007. Color codes: p-value greater than 0.1; p-value between 0.1 and 0.05; and p-value between 0.05 and 0.01.The blocks colored denote own-connections which are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 3.2 with $B = 1000$.

Table 8: Adjusted global stock market return connectedness obtained at the 1% FWER level.

	From																			
To	US	UK	FR	DE	HK	JP	AU	SG	ID	KR	PH	TW	TH	MY	AR	BR	CL	MX	TR	FIX
US	1.49	0.63	0.59	0.62	0.24	0.16	0.36	0.25	–	–	–	–	–	–	0.18	0.20	0.19	0.35	–	3.77
UK	0.61	1.45	0.86	0.77	0.33	0.18	0.35	0.26	–	–	–	–	–	–	0.16	–	–	0.28	–	3.81
FR	0.55	0.82	1.39	0.94	0.26	0.20	0.30	0.21	–	–	–	–	–	–	0.18	–	0.14	0.27	–	3.87
DE	0.61	0.77	0.98	1.45	0.30	0.18	0.30	0.25	–	–	–	–	–	–	–	0.15	–	0.28	–	3.82
HK	0.25	0.35	0.29	0.31	1.44	0.14	0.34	0.56	0.18	0.22	0.22	0.16	0.23	0.22	–	–	0.14	0.22	–	3.82
JP	0.27	0.31	0.34	0.31	0.22	2.28	0.34	0.27	0.14	0.27	–	0.17	–	–	–	0.17	–	0.17	–	2.99
AU	0.41	0.41	0.38	0.36	0.36	0.23	1.55	0.25	–	0.19	0.16	–	0.14	–	0.17	0.18	0.18	0.29	–	3.71
SG	0.25	0.27	0.22	0.25	0.54	0.16	0.23	1.36	0.25	0.24	0.32	0.18	0.34	0.33	0.14	–	–	0.18	–	3.90
ID	–	–	–	0.15	0.29	0.15	0.17	0.43	2.35	0.23	0.44	–	0.46	0.37	–	–	0.21	–	–	2.92
KR	0.19	0.22	0.22	0.23	0.34	0.26	0.27	0.38	0.19	2.22	–	0.21	0.37	–	–	–	–	0.17	–	3.05
PH	0.19	0.17	0.13	0.15	0.34	–	0.22	0.46	0.37	–	1.91	0.15	0.40	0.28	0.16	–	0.13	0.22	–	3.36
TW	0.17	0.15	0.19	0.25	0.32	0.19	0.14	0.36	–	0.24	0.18	2.55	0.17	0.17	–	–	–	0.19	–	2.72
TH	–	0.17	–	0.17	0.33	–	0.19	0.51	0.39	0.34	0.39	–	2.12	0.34	–	–	0.15	0.18	–	3.15
MY	–	–	–	0.16	0.39	–	0.19	0.59	0.42	–	0.34	0.15	0.44	2.40	–	–	–	0.18	–	2.86
AR	0.28	0.27	0.30	0.23	0.16	–	0.24	0.23	–	–	–	–	–	–	2.33	0.43	0.28	0.51	–	2.93
BR	0.31	0.24	0.23	0.26	0.16	0.18	0.29	–	–	–	–	–	–	–	0.44	2.36	0.37	0.40	–	2.90
CL	0.28	0.21	0.22	0.20	0.20	–	0.25	0.23	0.19	–	–	–	0.18	–	0.30	0.37	2.28	0.35	–	2.99
MX	0.43	0.36	0.36	0.35	0.28	–	0.31	0.24	–	–	0.18	–	–	–	0.38	0.31	0.23	1.83	–	3.43
TR	–	0.23	0.17	0.20	–	–	0.16	–	–	–	–	–	–	–	–	0.17	0.15	–	4.17	1.10
TIX	4.81	5.56	5.48	5.92	5.05	2.04	4.65	5.46	2.12	1.73	2.25	1.02	2.72	1.72	2.11	2.00	2.19	4.26	–	61.08

Notes: Sample period 10/01/1992 – 23/11/2007. Color codes: p-value greater than 0.1; p-value between 0.1 and 0.05; and p-value between 0.05 and 0.01. The blocks colored denote own-connections which are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 3.2 with $B = 1000$. Spillovers that are insignificant at the 1% FWER level are excluded (replaced with a dash), which necessitates renormalization of the spillover matrix to restore the matrix grand sum to 100%. The TIX, FIX and AIX are obtained from the renormalized values.

Table 9: Adjusted global stock market return connectedness at the 1% FWER level conditional on the four groups

	From																			
To	US	UK	FR	DE	HK	JP	AU	SG	ID	KR	PH	TW	TH	MY	AR	BR	CL	MX	TR	FIX
US	1.42	0.60	0.56	0.59	0.23	0.16	0.34	0.24	–	0.12	0.12	–	–	–	0.17	0.19	0.18	0.34	–	1.12
UK	0.56	1.32	0.78	0.70	0.30	0.16	0.32	0.24	–	0.12	0.11	–	–	–	0.15	0.12	0.12	0.26	–	0.88
FR	0.53	0.78	1.33	0.89	0.25	0.19	0.28	0.20	–	0.13	–	–	–	–	0.17	0.13	0.14	0.26	–	0.82
DE	0.56	0.71	0.91	1.34	0.28	0.17	0.28	0.23	–	0.13	–	–	–	–	0.13	0.14	0.12	0.26	–	0.79
HK	0.24	0.33	0.28	0.30	1.39	0.13	0.32	0.54	0.17	0.21	0.22	0.15	0.22	0.22	0.11	0.10	0.13	0.21	–	1.73
JP	0.26	0.29	0.33	0.30	0.21	2.19	0.32	0.26	0.13	0.26	–	0.17	–	0.11	0.10	0.17	–	0.17	–	1.10
AU	0.40	0.39	0.36	0.35	0.34	0.22	1.49	0.24	0.10	0.18	0.16	–	0.13	0.12	0.16	0.18	0.18	0.28	–	1.47
SG	0.24	0.27	0.22	0.24	0.52	0.15	0.22	1.33	0.25	0.23	0.31	0.18	0.33	0.32	0.13	–	0.12	0.18	–	2.06
ID	0.12	–	0.11	0.14	0.26	0.13	0.16	0.39	2.11	0.21	0.40	0.08	0.41	0.33	–	0.11	0.19	0.13	–	1.72
KR	0.18	0.20	0.20	0.21	0.31	0.24	0.25	0.35	0.17	2.04	0.10	0.19	0.34	0.11	0.11	–	0.11	0.16	–	2.30
PH	0.18	0.16	0.13	0.14	0.32	–	0.21	0.44	0.36	0.10	1.83	0.14	0.38	0.27	0.15	0.10	0.13	0.21	–	2.18
TW	0.16	0.14	0.18	0.23	0.30	0.18	0.14	0.34	0.09	0.23	0.17	2.40	0.16	0.16	0.11	–	0.10	0.18	–	2.06
TH	0.12	0.15	0.12	0.15	0.29	–	0.17	0.46	0.34	0.30	0.35	0.12	1.88	0.30	0.11	0.11	0.13	0.16	–	1.96
MY	0.10	0.12	–	0.14	0.35	0.11	0.17	0.52	0.37	0.12	0.30	0.13	0.39	2.14	–	–	0.13	0.16	–	1.80
AR	0.26	0.25	0.28	0.22	0.15	–	0.22	0.21	–	0.11	0.13	–	0.11	–	2.18	0.40	0.26	0.48	–	1.94
BR	0.29	0.22	0.22	0.24	0.15	0.17	0.27	0.12	–	0.11	–	–	0.13	–	0.41	2.20	0.35	0.38	–	1.92
CL	0.27	0.20	0.21	0.19	0.19	–	0.23	0.21	0.18	0.11	0.12	–	0.17	0.11	0.28	0.34	2.13	0.33	–	2.18
MX	0.39	0.33	0.33	0.32	0.25	0.12	0.28	0.22	–	0.13	0.16	–	0.12	0.12	0.35	0.29	0.21	1.66	–	2.77
TR	0.12	0.21	0.16	0.19	–	–	0.15	–	–	0.12	–	–	0.12	–	–	0.16	0.14	–	3.89	1.38
TIX	2.17	1.98	1.94	2.17	2.57	0.95	2.25	3.25	0.82	1.96	1.33	0.50	1.34	1.00	1.60	1.50	1.93	2.94	–	32.19

Notes: Sample period 10/01/1992 – 23/11/2007. Color codes: p-value greater than 0.1; p-value between 0.1 and 0.05; and p-value between 0.05 and 0.01. The blocks colored denote groups whose intra-group connections are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 4.1 with $B = 1000$. Spillovers that are insignificant at the 1% FWER level are excluded (replaced with a dash), which necessitates renormalization of the spillover matrix to restore the matrix grand sum to 100%.

Table 10: Adjusted global stock market return connectedness at the 1% FWER level among the four groups.

To	From				FIX
	Advanced Markets	Asian Markets	Latin America	Turkey	
Advanced Markets	30.77	6.30	5.03	–	11.33
Asian Markets	9.57	19.15	2.88	–	12.44
Latin America	6.86	2.45	11.68	–	9.32
Turkey	0.95	0.42	0.43	3.50	1.81
TIX	17.38	9.18	8.34	–	34.90

Notes: Sample period 10/01/1992 – 23/11/2007. Color code: [pink] p-value between 0.05 and 0.01. The blocks colored [blue] denote the intra-group connections which are not tested. The significance levels denote the step-down adjusted p-values for directional connectedness at the group level (30) with the hypothesis (31) imposing the structure of four groups. Spillovers that are insignificant at the 1% FWER level are excluded (replaced with a dash), which necessitates renormalization of the spillover matrix to restore the matrix grand sum to 100%.

Table 11: Link creation and deletion in the GFC and COVID subsamples.

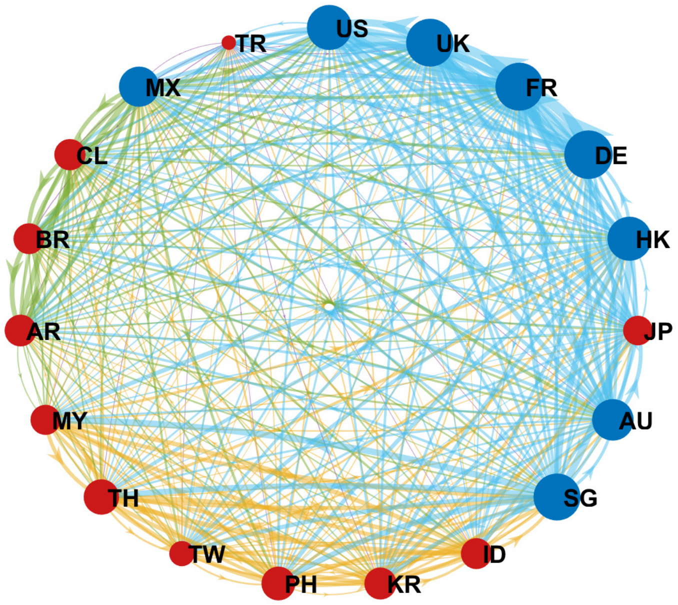
Period	Pre-GFC	GFC	Pre-COVID	COVID
Sample size	1034	627	770	865
# edges (sig)	78	74	58	99
% edges (sig)	25.5%	24.2%	19.0%	32.4%
AIX (raw)	40.97	70.91	41.68	60.78
AIX (adjusted)	30.47	50.49	26.89	48.73

	GFC changes	COVID changes
Link creation	31	55
From NA/EU	23	31
From APAC	–	10
From LATAM	8	12
From OTHER	–	2
GFEVD creation	19.41	20.39
From NA/EU	14.87	12.23
From APAC	–	3.32
From LATAM	4.54	4.23
From OTHER	–	0.61
Link deletion	35	14
From NA/EU	10	5
From APAC	5	3
From LATAM	3	3
From OTHER	17	3
GFEVD deletion	9.29	4.47
From NA/EU	2.50	1.76
From APAC	1.16	1.05
From LATAM	1.05	0.81
From OTHER	4.57	0.85

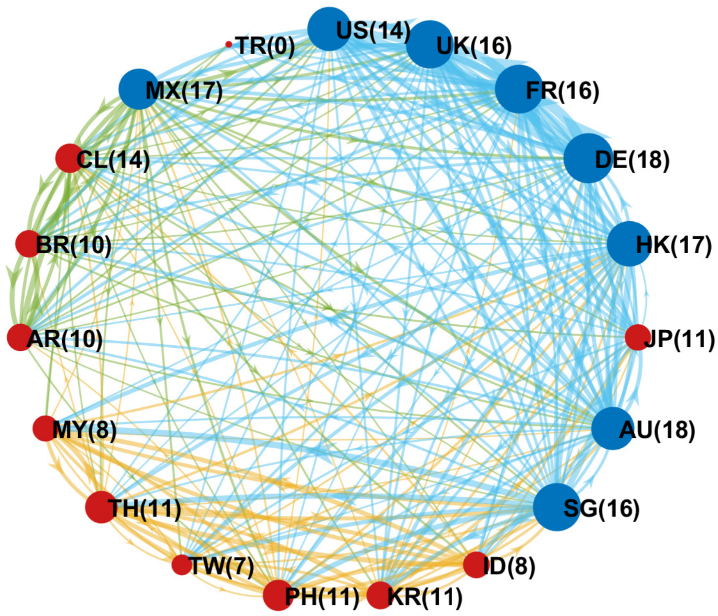
Notes: The number of significant edges and the percentage of significant edges use the significance level $\alpha = 1\%$. The percentage of significant edges is calculated using the number of significant edges divided by the total number of pairwise edges (306) in the 18-node network. AIX (raw) is calculated from all pairwise connections, irrespective of their statistical significance (i.e., it is the value that would be reported using the standard DY method). AIX (adjusted) is calculated from the renormalized pairwise connections that are significant at the 1% level, after removing the insignificant connections.

Figure 1: Global stock market return connectedness

(a) Raw network



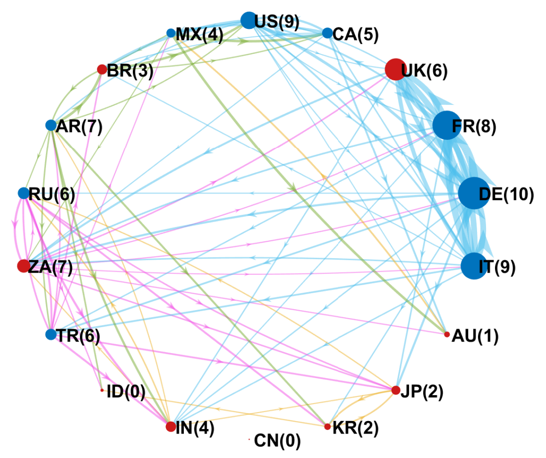
(b) Adjusted network obtained at the 1% FWER level



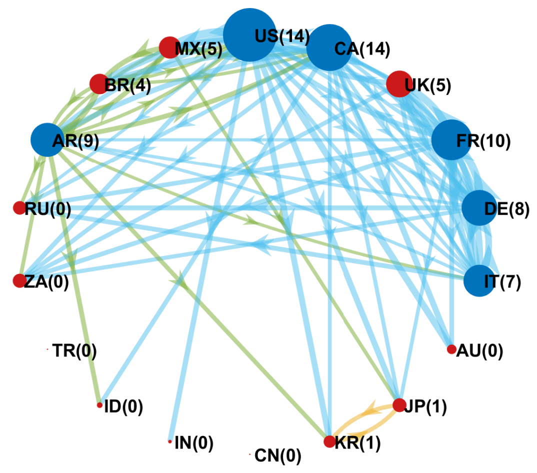
Notes: To obtain the results reported in panel (b), pairwise connections that are statistically insignificant at the 1% FWER level are set to zero. We then apply grand-sum normalization to ensure that the elements of the spillover table once again sum to 100%. These re-normalized spillover statistics are plotted in the figure. The number of significant outgoing edges for each node is reported in parentheses in panel (b).

Figure 2: Significant bilateral spillovers at the 1% FWER level in the global stock market volatility network before and after the GFC

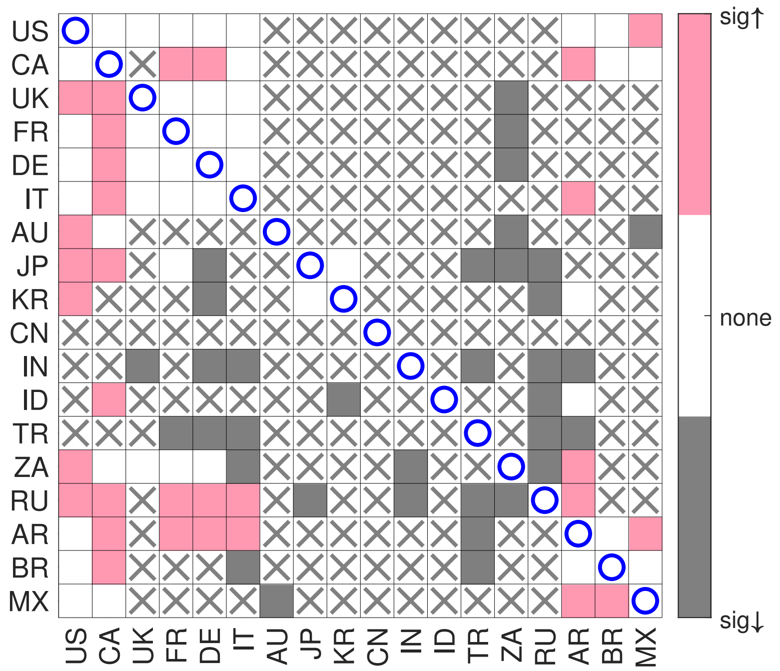
(a) Benchmark period, Jul 25, 2003–Jul 25, 2007



(b) Crisis period, Jul 26, 2007–Dec 31, 2009



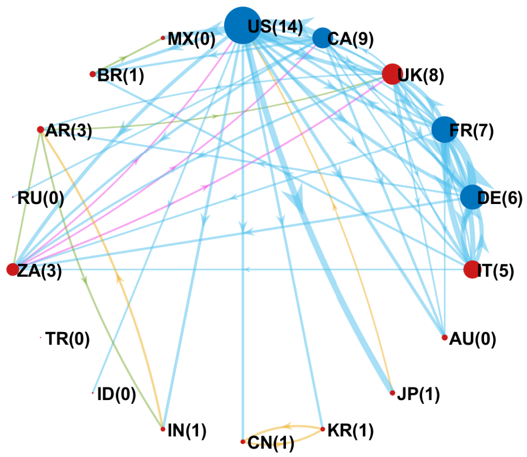
(c) Significance-transition map comparing the benchmark and crisis periods



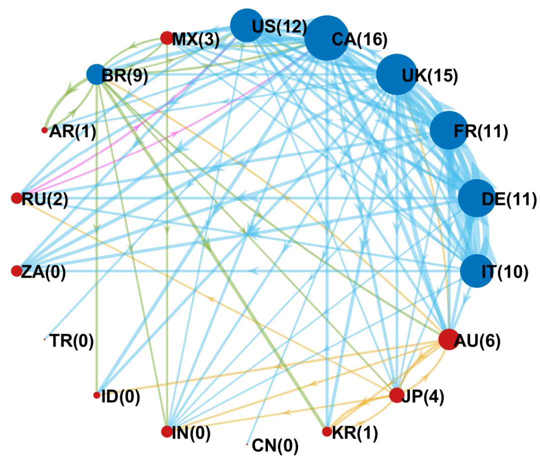
Notes: Panels (a) and (b) depict the network graphs for the pre-GFC and GFC periods, respectively. Insignificant edges at the 1% level have been removed, and the remaining edges have been renormalized so that the total add up to 100%. Blue (red) nodes represent markets with NIX >0 (NIX <0). The size of the nodes is proportional to TIX+FIX. Panel (c) is a significance-transition map that visualizes the changes in statistical significance at the 1% level for all edges between the two sample periods. Pink denotes edges that are insignificant in the pre-GFC period but become significant during the GFC. Grey denotes edges that are significant in the pre-GFC period but become insignificant during the GFC. White denotes edges that are significant pre-GFC and remain so. 'X' marks edges that are insignificant pre-GFC and remain so.

Figure 3: Significant bilateral spillovers at the 1% FWER level in the global stock market volatility network before and after the COVID crisis

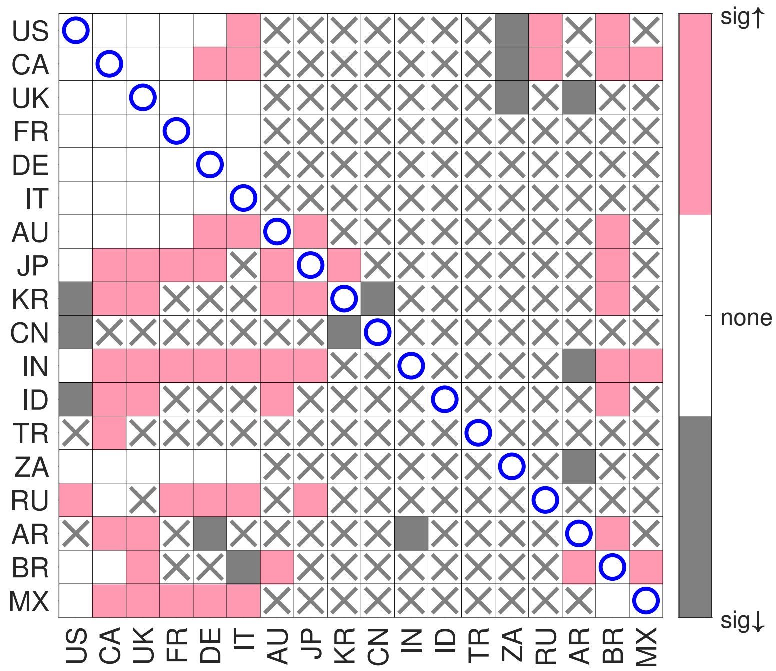
(a) Benchmark period, Jan 2, 2017–Dec 30, 2019



(b) Crisis period, Dec 31, 2019–May 5, 2023



(c) Significance-transition map comparing the benchmark and crisis periods



Notes: Panels (a) and (b) depict the network graphs for the pre-COVID and COVID periods, respectively. Insignificant edges at the 1% level have been removed, and the remaining edges have been renormalized so that the total add up to 100%. Blue (red) nodes represent markets with NIX > 0 (NIX < 0). The size of the nodes is proportional to TIX+FIX. is a significance-transition map that visualizes the changes in statistical significance at the 1% level for all edges between the two sample periods. Pink denotes edges that are insignificant in the pre-COVID period but become significant during the COVID period. Grey denotes edges that are significant in the pre-COVID period but become insignificant during the COVID period. White denotes edges that are significant pre-COVID and remain so. 'X' marks edges that are insignificant pre-COVID and remain so.

Online appendix

A Proofs

Proof of Proposition 3.1. Under $\mathcal{H}^*_{\bullet \not\leftarrow \bullet}$ and Assumption 1, the restricted innovations are independent across equations and i.i.d. over time within each equation. Hence all within-equation permutations of the restricted innovations are equally likely. Because $\widetilde{\mathbf{Y}}$ is generated from the same initial conditions and the same fixed autoregressive matrices $\mathbf{D}^*_l$ as the observed sample under $\mathcal{H}^*_{\bullet \not\leftarrow \bullet}$, but with within-equation permuted innovations, we have $\mathbf{Y} \overset{d}{=} \widetilde{\mathbf{Y}}$. The connectedness estimates are deterministic measurable functions of the sample. Therefore, by Theorem 1.3.7 of Randles and Wolfe (1979), $(\widehat{C}^H_\ell : \ell \in \mathcal{M}) \overset{d}{=} (\widetilde{C}^H_\ell : \ell \in \mathcal{M})$, and hence $\max_{\ell \in \mathcal{M}} \widehat{C}^H_\ell \overset{d}{=} \max_{\ell \in \mathcal{M}} \widetilde{C}^H_\ell$. □

Proof of Proposition 3.2. Proposition 3.1 implies that the random variables $\tilde{m}_1, \ldots, \tilde{m}_{B-1}, \hat{m}$ are exchangeable under $\mathcal{H}^*_{\bullet \not\leftarrow \bullet}$. This means that $\Pr\big(R(\hat{m}) = b\big) = 1/B$, for $b = 1, \ldots, B$, when $\mathcal{H}^*_{\bullet \not\leftarrow \bullet}$ holds true, since the ranks of B exchangeable random variables are uniformly distributed over $\{1, \ldots, B\}$. The stated result then follows on applying Lemma 1 in Romano and Wolf (2005) or Proposition 2.2 of Dufour (2006). □

Proof of Proposition 3.3. For the single-step procedure, rejection of at least one true null hypothesis is equivalent to

$$\min_{\ell \in \mathcal{M}_0} \tilde{p}_{\mathrm{SS}}(\widehat{C}^H_\ell) \leq \alpha.$$

Hence, for any non-empty $\mathcal{M}_0 \subseteq \mathcal{M}$, Assumption 2 with $\mathcal{P} = \mathrm{SS}$ gives

$$\Pr\left(\min_{\ell \in \mathcal{M}_0} \tilde{p}_{\mathrm{SS}}(\widehat{C}^H_\ell) \leq \alpha \;\middle|\; \bigcap_{\ell \in \mathcal{M}_0} \mathcal{H}_\ell \right) \leq \Pr\left(\min_{\ell \in \mathcal{M}} \tilde{p}_{\mathrm{SS}}(\widehat{C}^H_\ell) \leq \alpha \;\middle|\; \mathcal{H}^*_{\bullet \not\leftarrow \bullet} \right).$$

Under $\mathcal{H}^*_{\bullet \not\leftarrow \bullet}$, the minimum single-step adjusted p-value is attained by the largest observed statistic $\hat{m} = \max_{\ell \in \mathcal{M}} \widehat{C}^H_\ell$. Thus

$$\left\{ \min_{\ell \in \mathcal{M}} \tilde{p}_{\mathrm{SS}}(\widehat{C}^H_\ell) \leq \alpha \right\} = \{ \tilde{p}(\hat{m}) \leq \alpha \}.$$

By Proposition 3.2, with αB an integer and no ties,

$$\Pr(\tilde{p}(\hat{m}) \leq \alpha \mid \mathcal{H}^*_{\bullet \not\leftarrow \bullet}) = \alpha.$$

Therefore,

$$\Pr\left(\text{reject at least one } \mathcal{H}_\ell,\ \ell \in \mathcal{M}_0 \,\middle|\, \bigcap_{\ell \in \mathcal{M}_0} \mathcal{H}_\ell\right) \leq \alpha.$$

Since this holds for every non-empty $\mathcal{M}_0 \subseteq \mathcal{M}$, the procedure controls the finite-sample FWER at level α in the strong sense. If $\mathcal{M}_0$ is empty, the FWER is zero by definition. □

Proof of Proposition 3.4. For the step-down procedure, rejection of at least one true null hypothesis is equivalent to

$$\min_{\ell \in \mathcal{M}_0} \tilde{p}_{\mathrm{SD}}(\widehat{C}_\ell^H) \leq \alpha.$$

Hence, for any non-empty $\mathcal{M}_0 \subseteq \mathcal{M}$, Assumption 2 with $\mathcal{P} = \mathrm{SD}$ gives

$$\Pr\left(\min_{\ell \in \mathcal{M}_0} \tilde{p}_{\mathrm{SD}}(\widehat{C}_\ell^H) \leq \alpha \,\middle|\, \bigcap_{\ell \in \mathcal{M}_0} \mathcal{H}_\ell\right) \leq \Pr\left(\min_{\ell \in \mathcal{M}} \tilde{p}_{\mathrm{SD}}(\widehat{C}_\ell^H) \leq \alpha \,\middle|\, \mathcal{H}^*_{\bullet \not\leftrightarrow \bullet}\right).$$

By the ordering $\widehat{C}^H_{\pi_1} \geq \cdots \geq \widehat{C}^H_{\pi_M}$ and the monotonicity enforcement in Step 7 of Algorithm 3.2, the minimum step-down adjusted p-value over the full family is attained at the largest observed statistic:

$$\min_{\ell \in \mathcal{M}} \tilde{p}_{\mathrm{SD}}(\widehat{C}_\ell^H) = \tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_1}).$$

Moreover, the first step-down adjusted p-value is based on the full-family maximum and therefore coincides with the MC max-test p-value for $\hat{m} = \max_{\ell \in \mathcal{M}} \widehat{C}_\ell^H$:

$$\tilde{p}_{\mathrm{SD}}(\widehat{C}^H_{\pi_1}) = \tilde{p}(\hat{m}).$$

Thus,

$$\Pr\left(\min_{\ell \in \mathcal{M}} \tilde{p}_{\mathrm{SD}}(\widehat{C}_\ell^H) \leq \alpha \,\middle|\, \mathcal{H}^*_{\bullet \not\leftrightarrow \bullet}\right) = \Pr\left(\tilde{p}(\hat{m}) \leq \alpha \mid \mathcal{H}^*_{\bullet \not\leftrightarrow \bullet}\right).$$

By Proposition 3.2, with αB an integer and no ties,

$$\Pr\left(\tilde{p}(\hat{m}) \leq \alpha \mid \mathcal{H}^*_{\bullet \not\leftrightarrow \bullet}\right) = \alpha.$$

Therefore,

$$\Pr\left(\text{reject at least one } \mathcal{H}_\ell,\ \ell \in \mathcal{M}_0 \,\middle|\, \bigcap_{\ell \in \mathcal{M}_0} \mathcal{H}_\ell\right) \leq \alpha.$$

Since this holds for every non-empty $\mathcal{M}_0 \subseteq \mathcal{M}$, the procedure controls the finite-sample FWER at level α in the strong sense. If $\mathcal{M}_0$ is empty, the FWER is zero by definition. □

Proof of Proposition 4.1. Under $\mathcal{H}^*_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet}$ and Assumption 3, the group-specific restricted innovation blocks

$$\boldsymbol{\varepsilon}_t^{g*} = \mathbf{y}_t^g - \sum_{l=1}^{p} \mathbf{B}_l^{g*} \mathbf{y}_{t-l}^g, \qquad g = 1, \ldots, G,$$

form, for each group g, an i.i.d. sequence over time and are independent across groups at each t. Hence, for each g, the sequence $(\boldsymbol{\varepsilon}_1^{g*}, \ldots, \boldsymbol{\varepsilon}_T^{g*})$ is exchangeable, so all $T!$ temporal permutations are equally likely; since the groups are independent, the G temporal permutations are jointly equally likely. The simulated sample $\widetilde{\mathbf{Y}}$ is generated from the same initial conditions and fixed block-diagonal autoregressive matrices $\mathbf{B}_l^{g*}$ as under $\mathcal{H}^*_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet}$, but with independently drawn temporal permutations applied separately to each group's innovation sequence. Therefore $\mathbf{Y} \overset{d}{=} \widetilde{\mathbf{Y}}$. Since the inter-group connectedness estimates are deterministic functions of the sample, $(\widehat{C}_k^H : k \in \mathcal{K}) \overset{d}{=} (\widetilde{C}_k^H : k \in \mathcal{K})$, and hence $\max_{k \in \mathcal{K}} \widehat{C}_k^H \overset{d}{=} \max_{k \in \mathcal{K}} \widetilde{C}_k^H$. The permutation distribution of the full-family maximum over $\mathcal{K}$ therefore provides an exact finite-sample reference distribution.

We now establish FWER control. For the step-down procedure, rejection of at least one true inter-group null hypothesis is equivalent to $\min_{k \in \mathcal{K}_0} \tilde{p}_{\mathrm{SD}}(\widehat{C}_k^H) \leq \alpha$. For any non-empty $\mathcal{K}_0 \subseteq \mathcal{K}$, Assumption 4 with $\mathcal{P} = \mathrm{SD}$ gives

$$\Pr\left(\min_{k \in \mathcal{K}_0} \tilde{p}_{\mathrm{SD}}(\widehat{C}_k^H) \leq \alpha \;\middle|\; \bigcap_{k \in \mathcal{K}_0} \mathcal{H}_k \right) \leq \Pr\left(\min_{k \in \mathcal{K}} \tilde{p}_{\mathrm{SD}}(\widehat{C}_k^H) \leq \alpha \;\middle|\; \mathcal{H}^*_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet} \right).$$

By the ordering $\widehat{C}_{\pi_1}^H \geq \cdots \geq \widehat{C}_{\pi_K}^H$ and the monotonicity enforcement in Algorithm 4.1, the minimum step-down adjusted p-value over the full inter-group family satisfies

$$\min_{k \in \mathcal{K}} \tilde{p}_{\mathrm{SD}}(\widehat{C}_k^H) = \tilde{p}_{\mathrm{SD}}(\widehat{C}_{\pi_1}^H) = \tilde{p}(\hat{m}),$$

where $\hat{m} = \max_{k \in \mathcal{K}} \widehat{C}_k^H$ and the second equality holds because the first step-down adjusted p-value is based on the full-family maximum. By the exact finite-sample MC rank construction, with αB an integer and no ties,

$$\Pr\big(\tilde{p}(\hat{m}) \leq \alpha \;\big|\; \mathcal{H}^*_{\mathcal{G}_\bullet \not\leftarrow \mathcal{G}_\bullet}\big) = \alpha.$$

Therefore,

$$\Pr\left(\text{reject at least one } \mathcal{H}_k,\ k \in \mathcal{K}_0 \;\middle|\; \bigcap_{k \in \mathcal{K}_0} \mathcal{H}_k \right) \leq \alpha.$$

Since this holds for every non-empty $\mathcal{K}_0 \subseteq \mathcal{K}$, the procedure controls the finite-sample FWER at level α in the strong sense. If $\mathcal{K}_0$ is empty, the FWER is zero by definition. □

B Additional results for the Diebold and Yilmaz (2009) network

Table B.1 shows the spillover matrix obtained having imposed the four groups defined in the main text prior to excluding insignificant edges. Note that the point estimates of the pairwise spillovers are identical to those shown in Table 7. However, the aggregate indices differ, as they are now calculated using only the pairwise connectedness measures between the nodes that are in different groups.

Table B.2 reports the group-level spillover effects obtained from aggregation of Table B.1, without excluding insignificant edges.

Table B.1: Global stock market return connectedness network using the GFEVD conditional on the four groups.

	From																			
To	US	UK	FR	DE	HK	JP	AU	SG	ID	KR	PH	TW	TH	MY	AR	BR	CL	MX	TR	FIX
US	1.34	0.56	0.53	0.56	0.21	0.15	0.32	0.23	0.07	0.11	0.12	0.06	0.07	0.06	0.16	0.18	0.17	0.32	0.04	1.36
UK	0.52	1.24	0.73	0.66	0.29	0.15	0.30	0.22	0.05	0.11	0.10	0.06	0.09	0.06	0.14	0.12	0.12	0.24	0.06	1.15
FR	0.49	0.72	1.23	0.83	0.23	0.18	0.26	0.18	0.06	0.12	0.08	0.07	0.08	0.05	0.16	0.12	0.13	0.24	0.05	1.14
DE	0.51	0.65	0.83	1.22	0.26	0.15	0.25	0.21	0.07	0.12	0.09	0.09	0.09	0.07	0.12	0.13	0.11	0.23	0.05	1.19
HK	0.24	0.33	0.27	0.30	1.38	0.13	0.32	0.54	0.17	0.21	0.21	0.15	0.22	0.21	0.10	0.10	0.13	0.21	0.02	1.75
JP	0.25	0.28	0.31	0.28	0.20	2.08	0.31	0.24	0.13	0.25	0.06	0.16	0.07	0.11	0.10	0.16	0.08	0.16	0.04	1.31
AU	0.39	0.38	0.35	0.34	0.33	0.22	1.45	0.23	0.09	0.18	0.15	0.09	0.13	0.12	0.16	0.17	0.17	0.27	0.06	1.58
SG	0.24	0.26	0.21	0.24	0.51	0.15	0.22	1.31	0.24	0.23	0.31	0.17	0.33	0.32	0.13	0.07	0.12	0.17	0.03	2.12
ID	0.11	0.09	0.10	0.13	0.25	0.13	0.15	0.37	2.02	0.20	0.38	0.08	0.40	0.32	0.09	0.11	0.18	0.12	0.03	1.86
KR	0.17	0.19	0.20	0.20	0.30	0.23	0.24	0.34	0.17	1.98	0.10	0.19	0.33	0.11	0.10	0.09	0.10	0.15	0.06	2.39
PH	0.18	0.16	0.13	0.14	0.32	0.06	0.20	0.43	0.35	0.10	1.80	0.14	0.37	0.27	0.15	0.10	0.13	0.21	0.04	2.24
TW	0.15	0.14	0.18	0.23	0.29	0.18	0.13	0.33	0.09	0.23	0.17	2.36	0.15	0.16	0.10	0.05	0.10	0.17	0.05	2.12
TH	0.11	0.15	0.12	0.14	0.28	0.06	0.16	0.44	0.34	0.29	0.34	0.12	1.84	0.30	0.11	0.11	0.13	0.16	0.06	2.04
MY	0.10	0.11	0.09	0.14	0.33	0.10	0.17	0.50	0.36	0.12	0.29	0.13	0.37	2.05	0.09	0.03	0.12	0.16	0.01	1.95
AR	0.25	0.23	0.26	0.20	0.14	0.09	0.20	0.20	0.08	0.10	0.12	0.06	0.10	0.08	2.03	0.38	0.24	0.44	0.05	2.17
BR	0.28	0.21	0.21	0.23	0.14	0.16	0.25	0.11	0.09	0.10	0.09	0.04	0.12	0.03	0.39	2.07	0.33	0.35	0.07	2.13
CL	0.25	0.19	0.20	0.18	0.18	0.07	0.22	0.20	0.17	0.10	0.12	0.07	0.16	0.10	0.27	0.33	2.04	0.32	0.08	2.31
MX	0.37	0.31	0.31	0.31	0.24	0.11	0.27	0.21	0.08	0.12	0.16	0.09	0.12	0.12	0.33	0.27	0.20	1.59	0.03	2.86
TR	0.10	0.19	0.14	0.17	0.06	0.07	0.14	0.07	0.04	0.11	0.08	0.07	0.11	0.01	0.09	0.15	0.13	0.07	3.47	1.79
TIX	2.08	1.97	1.94	2.07	2.54	1.28	2.15	3.21	1.34	1.86	1.68	1.19	1.69	1.34	1.80	1.67	1.92	2.88	0.82	35.43

Notes: Sample period 10/01/1992 – 23/11/2007. Color codes: p-value greater than 0.1; p-value between 0.1 and 0.05; and p-value between 0.05 and 0.01. The blocks colored denote groups whose intra-group connections are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 4.1 with $B = 1000$.

Table B.2: Stock market return connectedness among the four groups.

To	From				FIX
	Advanced Markets	Asian Markets	Latin America	Turkey	
Advanced	30.52	6.25	4.99	0.35	11.58
Asian	9.49	18.99	2.85	0.24	12.59
Latin	6.81	2.44	11.58	0.23	9.47
Turkey	0.94	0.42	0.43	3.47	1.79
TIX	17.24	9.10	8.27	0.82	35.43

Notes: Sample period 10/01/1992 – 23/11/2007. Color code: p-value between 0.05 and 0.01. The blocks colored denote the intra-group connections which are not tested. The significance levels denote the step-down adjusted p-values for directional connectedness at the group level (30) with the hypothesis (31) imposing the structure of four groups.

C Additional results for the G20 volatility network

C.1 Details of the stock indices under consideration

In Table C.1, we provide details of the benchmark stock indices that we consider for each sovereign state in the G20, including their trading hours, which we report in local time. Note that we exclude both the European Union and African Union from our analysis, as they are unions of sovereign states rather than states in their own right. The asterisks denote cases of interest. First, note that the Indonesian Stock Exchange changed its regular trading hours on several occasions during our sample period and implemented different lunch break times on Fridays. Second, note that the Saudi Stock Exchange changed its trading days from Saturday–Wednesday to Sunday–Thursday in 2013. Due to the practice of weekend trading, we exclude Saudi Arabia from our analysis.

Table C.1: Stock market indices for the G20 sovereigns

Market	**Market Index**	**Trading Hours**
US	S&P 500 Index	9:30AM to 4:00PM
Canada	S&P/TSX Composite Index	9:30AM to 4:00PM
UK	FTSE 100 Index	8:00AM to 4:30PM
France	CAC 40 Index	9:00AM to 5:30PM
Germany	DAX Index	9:00AM to 5:30PM
Italy	FTSE MIB Index	9:00AM to 5:30PM
Australia	ASX All Ordinaries Index	10:00AM to 4:00PM
Japan	Nikkei 225 Index	9:00AM to 11:30AM, 12:30PM to 3:00PM
South Korea	Korea SE KOSPI Index	9:00AM to 3:30PM
China	Shanghai SE Composite Index	9:30AM to 11:30AM, 1:00PM to 3:00PM
India	S&P BSE SENSEX Index	9:15AM to 3:30PM
Indonesia	Jakarta SE Composite Index	8:30AM to 5:00PM*
Saudi Arabia	Tadawul All Share Index	10:00AM to 3:00PM*
Turkey	Borsa Istanbul 100 Index	9:20AM to 6:15PM
South Afirca	FTSE/JSE Top 40 Index	9:00AM to 5:00PM
Russia	MOEX Russia Index	10:00AM to 6:40PM
Argentina	S&P Merval Index	11:00AM to 6:00PM
Brazil	Bovespa Index	10:00AM to 5:55PM
Mexico	S&P IPC Index	8:30AM to 3:00PM

C.2 The G20 volatility network prior to and during the GFC

In Tables C.2 and C.3, we report the connectedness matrices obtained at the 1% FWER level via our step-down multiple testing procedure in the pre-GFC period (July 25, 2003 to July 25, 2007) and the GFC period (July 26, 2007 to December 31, 2009). Edges that are insignificant at the 1% FWER level are excluded from the tables (as indicated by the dashes) and a further normalization step is then applied to restore the matrix grand sum to 100%. The values recorded in Tables C.2 and C.3 are used to draw Figure 2 and as inputs in the construction of Table 11 in the manuscript.

Table C.2 reveals a high level of connectedness among the US, UK, France, Germany, and Italy in the pre-GFC period. Canada experiences significant bilateral volatility spillover with respect the US—its largest trading partner and a country with which it shares close economic, financial and geopolitical ties—but less so with the European markets. There is evidence of smaller clusters among other markets, such as between Japan and Korea, and among the Latin American markets. Both the TIX and FIX for China are insignificant at the 1% FWER level in the pre-GFC period, indicating its relative isolation within the global volatility network. The adjusted AIX of 30.47% indicates that roughly one-third of system-wide connectedness can be attributed to bilateral spillovers, with the remaining two-thirds due to within-market effects. By contrast, the raw AIX value obtained from the DY method without inference is much higher, at approximately 41%.

Table C.3 summarizes spillover activity during the GFC period. Several changes in the topology of the volatility network can be seen in Table C.3. First, there is a marked intensification of aggregate spillover activity. The adjusted AIX has increased to 50.49% (and the raw AIX to 71%). This is largely due to an increase in the number and strength of spillovers originating from the North American and European markets. By contrast, volatility spillovers from Australia, India, Turkey, South Africa and Russia diminish to the point of insignificance. The negligible volatility connectedness of China and Indonesia with the other markets is maintained.

C.3 The G20 volatility network prior to and during COVID

In Tables C.4 and C.5, we report the connectedness matrices obtained at the 1% FWER level via our step-down multiple testing procedure in the pre-COVID period (January 2, 2017 to December 30, 2019) and the COVID period (December 31, 2019 to May 5, 2023). Inference renormalization proceeds as in the previous subsection. The values recorded in Tables C.4 and C.5 are used to draw Figure 3 and as inputs in the construction of Table 11 in the manuscript.

Table C.4 and C.5 reveal that the topology of the volatility network pre-COVID is similar to that observed pre-GFC to the extent that the majority of spillover activity is driven by the advanced markets in North America and Europe, with little spillover activity originating elsewhere. The adjusted AIX of 26.89% is slightly lower than in the pre-GFC period, while the raw AIX is 41.68%.

Table C.2: G20 stock market log-volatility connectedness network using the GFEVD, July 25, 2003 to July 25, 2007.

	From																		FIX
To	US	CA	UK	FR	DE	IT	AU	JP	KR	CN	IN	ID	TR	ZA	RU	AR	BR	MX	
US	3.30	0.36	0.23	0.31	0.46	0.30	–	–	–	–	–	–	–	–	–	0.20	0.40	–	2.26
CA	0.53	4.18	–	–	–	0.23	–	–	–	–	–	–	–	–	–	–	0.20	0.41	1.38
UK	–	–	2.55	0.97	1.02	0.75	–	–	–	–	–	–	–	0.26	–	–	–	–	3.00
FR	0.18	–	0.76	2.12	1.34	0.97	–	–	–	–	–	–	–	0.19	–	–	–	–	3.44
DE	0.25	–	0.71	1.24	2.20	0.97	–	–	–	–	–	–	–	0.19	–	–	–	–	3.36
IT	0.19	–	0.68	1.14	1.17	2.37	–	–	–	–	–	–	–	–	–	–	–	–	3.18
AU	–	0.32	–	–	–	–	4.61	–	–	–	–	–	–	0.19	–	–	–	0.43	0.95
JP	–	–	–	0.26	0.36	–	–	3.87	0.26	–	–	–	0.29	0.25	0.26	–	–	–	1.69
KR	–	–	–	–	0.21	–	–	0.27	4.42	–	–	–	–	–	0.32	0.33	–	–	1.13
CN	–	–	–	–	–	–	–	–	–	5.56	–	–	–	–	–	–	–	–	–
IN	–	–	0.23	–	0.23	0.24	–	–	–	–	3.77	–	0.34	–	0.37	0.38	–	–	1.79
ID	–	–	–	–	–	–	–	–	0.19	–	–	4.89	–	–	0.21	0.26	–	–	0.67
TR	–	–	–	0.22	0.29	0.29	–	–	–	–	–	–	4.22	–	0.29	0.25	–	–	1.34
ZA	–	0.19	0.34	0.28	0.36	0.24	–	–	–	–	0.20	–	–	3.59	0.35	–	–	–	1.97
RU	–	–	–	–	–	–	–	0.22	–	–	0.26	–	0.33	0.22	4.53	–	–	–	1.02
AR	0.24	–	–	–	–	–	–	–	–	–	–	–	0.24	–	–	4.82	0.26	–	0.74
BR	0.52	–	–	–	–	0.19	–	–	–	–	–	–	0.26	–	–	0.45	3.93	0.21	1.62
MX	0.22	0.42	–	–	–	–	0.30	–	–	–	–	–	–	–	–	–	–	4.62	0.94
TIX	2.12	1.29	2.96	4.44	5.44	4.17	0.30	0.49	0.45	–	0.45	–	1.47	1.30	1.80	1.87	0.86	1.05	30.47

Notes: Sample period July 25, 2003 – July 25, 2007. Color codes: p-value greater than 0.1; p-value between 0.1 and 0.05; and p-value between 0.05 and 0.01. The blocks colored denote own-connections which are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 3.2 with $B = 1000$. Spillovers that are insignificant at the 1% FWER level are excluded (replaced with a dash), which necessitates renormalization of the spillover matrix to restore the matrix grand sum to 100%.

Table C.3: G20 stock market log-volatility connectedness network using the GFEVD, July 26, 2007 to December 31, 2009.

	From																		
To	US	CA	UK	FR	DE	IT	AU	JP	KR	CN	IN	ID	TR	ZA	RU	AR	BR	MX	FIX
US	1.37	0.65	0.43	0.56	0.58	0.46	–	–	–	–	–	–	–	–	–	0.50	0.46	0.53	4.18
CA	0.78	1.54	–	0.54	0.54	0.47	–	–	–	–	–	–	–	–	–	0.68	0.48	0.52	4.02
UK	0.75	0.60	1.54	1.04	0.90	0.74	–	–	–	–	–	–	–	–	–	–	–	–	4.02
FR	0.68	0.55	0.85	1.45	1.10	0.92	–	–	–	–	–	–	–	–	–	–	–	–	4.10
DE	0.66	0.54	0.74	1.10	1.58	0.94	–	–	–	–	–	–	–	–	–	–	–	–	3.97
IT	0.56	0.49	0.65	0.97	1.00	1.44	–	–	–	–	–	–	–	–	–	0.45	–	–	4.12
AU	1.07	0.95	–	–	–	–	3.54	–	–	–	–	–	–	–	–	–	–	–	2.02
JP	0.91	0.63	–	0.59	–	–	–	2.76	0.67	–	–	–	–	–	–	–	–	–	2.79
KR	0.93	–	–	–	–	–	–	0.68	3.27	–	–	–	–	–	–	0.68	–	–	2.29
CN	–	–	–	–	–	–	–	–	–	5.56	–	–	–	–	–	–	–	–	–
IN	–	–	–	–	–	–	–	–	–	–	5.56	–	–	–	–	–	–	–	–
ID	–	0.66	–	–	–	–	–	–	–	–	–	4.22	–	–	–	0.68	–	–	1.33
TR	–	–	–	–	–	–	–	–	–	–	–	–	5.56	–	–	–	–	–	–
ZA	0.68	0.62	0.62	0.65	0.58	–	–	–	–	–	–	–	–	1.85	–	0.55	–	–	3.71
RU	0.46	0.58	–	0.65	0.70	0.67	–	–	–	–	–	–	–	–	1.99	0.51	–	–	3.56
AR	0.65	0.76	–	0.45	0.47	0.46	–	–	–	–	–	–	–	–	–	1.76	0.55	0.47	3.80
BR	0.99	0.82	–	–	–	–	–	–	–	–	–	–	–	–	–	0.83	2.19	0.73	3.36
MX	1.12	0.75	–	–	–	–	–	–	–	–	–	–	–	–	–	0.69	0.65	2.34	3.22
AIX	10.24	8.58	3.30	6.53	5.86	4.66	–	0.68	0.67	–	–	–	–	–	–	5.59	2.14	2.25	50.49

Notes: Sample period July 26, 2007 to December 31, 2009. Color codes: p-value greater than 0.1; p-value between 0.1 and 0.05; and p-value between 0.05 and 0.01. The blocks colored denote own-connections which are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 3.2 with $B = 1000$. Spillovers that are insignificant at the 1% FWER level are excluded (replaced with a dash), which necessitates renormalization of the spillover matrix to restore the matrix grand sum to 100%.

Table C.4: G20 stock market log-volatility connectedness network using the GFEVD, January 2, 2017 to December 30, 2019.

	From																		
To	US	CA	UK	FR	DE	IT	AU	JP	KR	CN	IN	ID	TR	ZA	RU	AR	BR	MX	FIX
US	3.30	0.97	0.28	0.29	0.42	–	–	–	–	–	–	–	–	0.28	–	–	–	–	2.25
CA	1.22	3.48	0.27	0.31	–	–	–	–	–	–	–	–	–	0.27	–	–	–	–	2.07
UK	0.61	0.35	2.56	0.65	0.60	0.26	–	–	–	–	–	–	–	0.29	–	0.24	–	–	3.00
FR	0.45	0.32	0.56	2.25	1.20	0.78	–	–	–	–	–	–	–	–	–	–	–	–	3.31
DE	0.53	0.26	0.50	1.23	2.40	0.64	–	–	–	–	–	–	–	–	–	–	–	–	3.16
IT	0.38	0.27	0.26	0.93	0.75	2.98	–	–	–	–	–	–	–	–	–	–	–	–	2.58
AU	0.50	0.40	0.27	0.25	–	–	4.13	–	–	–	–	–	–	–	–	–	–	–	1.43
JP	0.97	–	–	–	–	–	–	4.59	–	–	–	–	–	–	–	–	–	–	0.97
KR	0.39	–	–	–	–	–	–	–	4.83	0.34	–	–	–	–	–	–	–	–	0.73
CN	0.40	–	–	–	–	–	–	–	0.34	4.81	–	–	–	–	–	–	–	–	0.74
IN	0.44	–	–	–	–	–	–	–	–	–	4.82	–	–	–	–	0.30	–	–	0.74
ID	0.30	–	–	–	–	–	–	–	–	–	–	5.26	–	–	–	–	–	–	0.30
TR	–	–	–	–	–	–	–	–	–	–	–	–	5.56	–	–	–	–	–	–
ZA	0.56	0.40	0.35	0.29	0.37	0.25	–	–	–	–	–	–	–	3.06	–	0.28	–	–	2.50
RU	–	0.28	–	–	–	–	–	–	–	–	–	–	–	–	5.28	–	–	–	0.28
AR	–	–	–	–	0.33	–	–	–	–	–	0.36	–	–	–	–	4.86	–	–	0.69
BR	0.49	0.36	–	–	–	0.35	–	–	–	–	–	–	–	–	–	–	4.35	–	1.20
MX	0.66	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0.30	4.60	0.96
AIX	7.90	3.60	2.49	3.95	3.66	2.28	–	–	0.34	0.34	0.36	–	–	0.85	–	0.81	0.30	–	26.89

Notes: Sample period January 2, 2017 to December 30, 2019. Color codes: p-value greater than 0.1; p-value between 0.1 and 0.05; and p-value between 0.05 and 0.01. The blocks colored denote own-connections which are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 3.2 with $B = 1000$. Spillovers that are insignificant at the 1% FWER level are excluded (replaced with a dash), which necessitates renormalization of the spillover matrix to restore the matrix grand sum to 100%.

Table C.5: G20 stock market log-volatility connectedness network using the GFEVD, December 31, 2019 to May 5, 2023.

	From																		FIX
To	US	CA	UK	FR	DE	IT	AU	JP	KR	CN	IN	ID	TR	ZA	RU	AR	BR	MX	
US	1.79	1.24	0.44	0.56	0.51	0.42	–	–	–	–	–	–	–	–	0.33	–	0.28	–	3.77
CA	0.99	1.80	0.49	0.49	0.49	0.37	–	–	–	–	–	–	–	–	0.28	–	0.36	0.28	3.75
UK	0.43	0.61	1.89	0.92	0.90	0.81	–	–	–	–	–	–	–	–	–	–	–	–	3.66
FR	0.51	0.58	0.77	1.53	1.16	1.00	–	–	–	–	–	–	–	–	–	–	–	–	4.03
DE	0.47	0.57	0.76	1.19	1.55	1.02	–	–	–	–	–	–	–	–	–	–	–	–	4.01
IT	0.43	0.50	0.77	1.13	1.13	1.59	–	–	–	–	–	–	–	–	–	–	–	–	3.97
AU	0.40	0.62	0.69	0.48	0.49	0.40	1.79	0.33	–	–	–	–	–	–	–	–	0.36	–	3.77
JP	0.30	0.48	0.42	0.30	0.31	–	0.40	2.75	0.30	–	–	–	–	–	–	–	0.29	–	2.80
KR	–	0.47	0.52	–	–	–	0.34	0.40	3.30	–	–	–	–	–	–	–	0.52	–	2.25
CN	–	–	–	–	–	–	–	–	–	5.56	–	–	–	–	–	–	–	–	–
IN	0.29	0.41	0.39	0.27	0.27	0.27	0.32	0.29	–	–	2.44	–	–	–	–	–	0.29	0.30	3.11
ID	–	0.53	0.46	–	–	–	0.35	–	–	–	–	3.86	–	–	–	–	0.36	–	1.70
TR	–	0.32	–	–	–	–	–	–	–	–	–	–	5.24	–	–	–	–	–	0.32
ZA	0.45	0.58	0.55	0.51	0.49	0.39	–	–	–	–	–	–	–	2.58	–	–	–	–	2.97
RU	0.42	0.47	–	0.45	0.42	0.36	–	0.27	–	–	–	–	–	–	3.17	–	–	–	2.39
AR	–	0.35	0.31	–	–	–	–	–	–	–	–	–	–	–	–	4.39	0.50	–	1.16
BR	0.41	0.60	0.37	–	–	–	0.32	–	–	–	–	–	–	–	–	0.35	3.19	0.32	2.36
MX	0.42	0.52	0.42	0.37	0.37	0.29	–	–	–	–	–	–	–	–	–	–	0.31	2.85	2.70
AIX	5.54	8.84	7.35	6.68	6.54	5.32	1.72	1.29	0.30	–	–	–	–	–	0.61	0.35	3.29	0.90	48.73

Notes: Sample period December 31, 2019 to May 5, 2023. Color codes: (red) p-value greater than 0.1; (light red) p-value between 0.1 and 0.05; and (pink) p-value between 0.05 and 0.01. The blocks colored (blue) denote own-connections which are not tested. The significance levels are obtained using the step-down adjusted p-values in Algorithm 3.2 with $B = 1000$. Spillovers that are insignificant at the 1% FWER level are excluded (replaced with a dash), which necessitates renormalization of the spillover matrix to restore the matrix grand sum to 100%.

Table C.5 shows the structure of the volatility network during the COVID period. The table reveals a marked increase in the number of statistically significant edges during the pandemic and the associated economic, financial and social turmoil. Most possible spillovers originating in the North American and European markets are now significant at the 1% FWER level, but newly significant spillovers originating in Australia, Japan, Russia, Brazil and Mexico are also observed. The only market in which widespread deletion of outward links is documented is South Africa. The adjusted AIX almost doubles, reaching 48.73%, while the raw AIX rises to 60.78%.